



**GOVERNMENT OF WEST BENGAL
PUBLIC WORKS (ROADS) DIRECTORATE
NATIONAL HIGHWAY WING**



**ADDENDA OF NEW ITEMS
TO
SCHEDULE OF RATES
2019-20
ROAD & BRIDGE WORKS
1 February 2020**

Foreword

NH schedule of rates 2019 -2020 was published with effect from 1st June 2019 based on the input received from different Manufacturing houses/Organisations/ Related authorities. Ministry's **Standard Data Book** for analysing of rates (1st revision, 2003) as well as the guideline of MoRTH's Specification (5th revision, August 2013) were followed.

As mentioned in the prefacing report to the NH SOR 2019-20 some new items have been framed by the NH Planning & Design Circle as an addenda to the NH SOR, 2019-20.

It is the Ministry's strong endeavour to introduce new technologies in the construction and maintenance of **National Highways** throughout for quite some time past. For this a good number circulars have also been published by the Ministry time to time. But it is to be appreciated that for not having any standardise norm for compilation of the rates for the new items by framing of its narration and consumption of materials, Labour as well as T& P involvement is highly complex and requires a lot of research works.

State PWD maintains a good number of steel bridges on different NHs. Since a long time back it has been felt by the Engineer officers who are to maintain steel bridges that there are no suitable item works in the NH SOR to help them. Due to want of standard data book support in this specific sector state Engineers are of the practice of framing items related to maintenance of structural steel bridges in a highly de-centralised manner and to get the item rate approved by the concerned Superintending Engineer. **This is for the first time** provision of items of works related to **construction as well as maintenance of structural steel bridges** have been made in the NH SOR, which would be of great help to those engineers.

Apart from adoption of new technology, items with environment friendly mode of road construction work have also been incorporated to protect our nature against pollution. The new items are cost effective as well as economical. As an efficient mode of waste management, new items like **Waste Plastic Modified Dense Graded Hot Bituminous mix** has been framed. In order to avoid environment pollution item like **Mechanised Mastic Asphalt** as been included.

To enhance the better safety to the NH users, several new items for road signage, markings have been added in the SOR. **For cost optimization in road construction and to achieve greater load bearing capacities** of the base and sub base layers as well as bituminous surfaces different items with geo-grid and geo-textile materials have been formulated.

The finalisation of items of work was not possible without active co-operation of Engineers working in NH Planning & Design Circle as well as Engineer Officers attached with this office. Hope this new corrigendum items will major help all the construction wing working under National Highway in West Bengal

All users of this Addenda are requested to feel free in suggesting anything for up gradation of the documents in future.

Date-28th January'2020

KHADYA BHABAN

1STFloor,Block-A, Room No.-103
11A,Mirza Ghalib Street
Kolkata-700 087
Phone: (033) 2252 0516
Fax: (033) 2252 0553
E-mail: cenhwb@gmail.com



(Saumitra Kumar Maiti)

Chief Engineer
(National Highways)

&

Chairman,
NH Schedule Review Committee
P.W. (Roads) Directorate
Government of West Bengal

Report

NH schedule of rates 2019 -2020 was published with effect from 1st June 2019 based on the input received from different Manufacturing houses/Organisations/ Related authorities. Ministry's Standard Data Book for analysing of rates (1st revision, 2003) as well as the guideline of MORTH Specification (5th revision, August 2013) were followed.

In the prefacing report of the NH schedule of rates 2019-2020, it was mentioned that some new items are in the pipeline and would be introduced very soon as an addendum to the NH SOR, 2019-20.

It is the Ministry's strong endeavour to introduce new technologies in the construction and maintenance of National Highways throughout for quite some time past. For this a good number circulars have also been published by the Ministry time to time. But it is to be appreciated that for not having any standardise norm for compilation of the rates for the new items by framing of its narration and consumption of materials, Labour as well as T& P involvement is highly complex and requires a lot of research works.

State PWD maintains a good number of steel bridges on different NHs. Since a long time back it has been felt by the Engineer officers who are to maintain steel bridges that there are no suitable item works in the NH SOR to help them. Due to want of standard data book support in this specific sector state PWD also remained silent. As such state Engineers are of the practice of framing items related to maintenance of structural steel bridges in a highly de-centralised manner and to get the item rate approved by the concerned Superintending Engineer. This is for the first time provision of items of works related to construction as well as maintenance of structural steel bridges have been made in the NH SOR. In this context, Indian Railways' unified schedule of Rates available on portal have been of much help.

Apart from adoption of new technology, items with environment friendly mode of road construction work have also been incorporated to protect our nature against pollution. The new items are cost effective as well as economical.

Following new items of works have been included in the chapter 17 as miscellaneous items: -

A. Related to Bituminous Base and Surface Courses works (Chapter-5 of NHSOR 19-20)

A-1. Mechanised Mastic Asphalt

A-2. Waste Plastic Modified Dense Graded Hot Bituminous mix

A-3. Water proofing work with the provision of Asphalt Seal pad on concrete surface.

B. Related to Cement Concrete Pavement works (Chapter-6 of NHSOR 19-20)

B-1. White Topping & Short Panel Concrete Pavement.

B-2. Concrete Sawing.

B-3. Concrete Paver Block.

C. Related to Geo-synthetics and Reinforced Earthworks (Chapter-7 of NHSOR 19-20)

- C-1. Ground improvement with Prefabricated Vertical Drain.
- C-2 & 3. Glass fibre reinforced Geo-grid.
- C-4, 5 & 6. High Modulus Bi-axial Geo-grid.
- C-7. High Modulus Tri-axial Geo-grid.
- C-8. High Strength flexible Geo-grid.
- C-9 & 10. Reinforced Geo-composite 3-dimensional Matrix(Woven).
- C-11. Woven Geo-Textile.
- C-12. Fiber glass geogrid from high quality glass fiber strands and coated with a polymer modified bitumen.
- C-13. Non-Woven Geotextile.
- C-14 & 15. Non-Woven Drainage Composite.
- C-16 to 42. Gabion Structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire Mesh Gabion Box.
- C-43 to 45. Cylindrical/Sack Gabion.

D. Related to Traffic Signs, Markings and Other Road Appurtenances (Chapter-8 of SOR)

- D-1. Raised Profile Edge Marking with Reflectorized Hot Applied Thermoplastic Compound on Bituminous Road Surface.
- D-2. Hybrid Solar-Retro Reflective Sign.
- D-3. Solar Powered Road Studs.
- D-4. Wire Rope Safety Cash Barrier.
- D-5. RPM with Shank.
- D-6. Median Marker.
- D-7. Retro-Reflective Studs.
- D-8. Rumble Marking.
- D-9. Audible Vibratory Profile Marking.

E. Related to Bridge structure

- E-1&2. Bridge Work Super Structure Steel.

- E-3. Metalizing of Steel work.
- E-4. Cleaning of Steelwork of Bridge.
- E-5, 6, 7 & 8. Painting on Bridge Structure.

The compilation/framing of 2nd Addenda was a robust task involving collection of data from different Manufacturing Houses, Construction Agencies, and Specialised Firms to fabricate the new items of work with consultation with Government officers working in NH construction wing. Also verification of item of works & rates in consultation with different expert Engineers and specialised agencies for finalising schedule items of work with cost of works was not possible without active co-operation of Engineers working in Planning & Design Circle(NH) as well as Engineer Officers attached with o/o the Chief Engineer (NH). Hope this new corrigendum items will major help all the construction wing working under National Highway in West Bengal. The contribution made by Sri Rajib Chattoraj, the then Assistant Chief Engineer-I (NH), in framing the narrative of each and every new items is praiseworthy. His subsequent contribution even after demitting that chair is being highly acclaimed by NH Wing.

Date: 28th January 2020

KHADYA BHABAN

2nd Floor, Block-A, Room No.-207
11A, MirzaGhalib Street
Kolkata-700 087
Phone & Fax: (033) 2252 1524
E-mail: sepwrndh@gmail.com



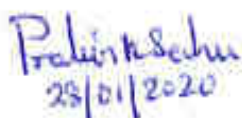
(Parthasarathi De)
Executive Engineer
NH Planning & Design Division-II
P.W(Roads).Directorate
Government of West Bengal



(Samaresh Bhattacharyya)
Superintending Engineer
NH Circle-I
P.W(Roads).Directorate
Government of West Bengal



(Sovan Kumar Mukhopadhyay)
Superintending Engineer,
NH Planning & Design Circle,
And
Member Secretary & Convener,
NH Schedule Review Committee
P.W. (Roads) Directorate
Government of West Bengal



(Prabir Kr. Sahu)
Assistant Chief Engineer-I
National Highways
P.W(Roads).Directorate
Government of West Bengal



(Saumitra Kumar Maiti)
Chief Engineer (NH)
P.W(Roads).Directorate
Government of West Bengal

CONTENTS

Page No.

GENERAL

Chapter -A

Chapter - A	-	Approximate material requirements for different Items of work	1-47
Chapter -17	-	Miscellaneous	
A.		New Items related to base and surface courses (Bituminous)	48
B.		New Items related to Cement concrete Pavement	49-50
C.		New Items related to Geo Synthetics and Reinforced Earth	51-63
D.		New Items related to Traffic Signs, Marking, and other Road appurtenances	64-66
E.		Steel Structure of Bridge	67-68
TABLES			
Table - I (Part-C)	-	Rate of Paver Block in Different District	69
ANNEXURE			
Annexure - III		Illustrative Examples	70-77

CHAPTER - A

APPROXIMATE MATERIAL REQUIREMENT FOR DIFFETENT ITEMS OF WORKS

CHAPTER -A

Items & materials

A. Consumption of Materials For New Items Related to Bases and Surface courses (Bituminous) Chapter 5 of NH SOR 2019-20

Item No. A-1. Mechanized Mastic Asphalt (25mm thick)

Unit- cum

Materials Requirement

Follow as per Chapter 'A' Item No. 5.14.of NH SOR 2019-20.

Item No. A-2. Waste Plastic Modified Dense Graded Hot Bituminous Mix For Wearing Course.

Unit- cum

Materials Requirement

Per cum of Waste Plastic Modified Dense Graded Hot Bituminous Mix For Wearing Course.

Size of crushed stone aggregates	Grading - II, 13.2 mm nominal max ^m aggregate size (layer thickness 30-40mm) Cum
20mm	-
13.2mm	0.07
10mm	0.24
5.6mm	0.25
2.36mm	0.18
1.18mm	0.13
600 micron	0.13
300 micron	0.13
150 micron	0.10
75 micron	0.23
Bitumen content (%) by mass of total mix. Bitumen may be VG 30 or VG 40 as per design mix depending on location of site and volume of traffic.	5.4 % (min.)
Bitumen content by mass	127 Kg (Min)

Items & materials

Waste Plastic content by mass @ 8% by wt of bitumen	10.16 kg
Filler(@ 2 % by total weight of aggregate)(Cement /lime* / Rock dust)	44.57 kg

Notes: -- Quantities are only for estimation. For execution, grading shall conform to table 500-17 of MoRT&H specification for road and bridge works (5th Revision ,2013)

* Lime as a filler material should be Hydrated Lime [i.e. chemically known as Ca(OH)_2]

Item No. A-3. Water proofing work with the provision of Asphalt Seal Pad for bridge deck, Road Over Bridges, Parking Decks, Bridge Ramps, Elevated Roadways, Pedestrian Subways (Concrete Surface) with 5mm thick Asphalt seal pad made of atactic polypropylene polymer (APP) modified bituminous pad reinforced with nonwoven polymer mat.

Unit- sqm

Materials Requirement

Heavy duty polymer modified bituminous Mastic Pad (APP) 5mm thick specially designed for bridge deck Road Over Bridges, Parking Decks, Bridge Ramps, Elevated Roadways, Pedestrian Subways, configured with high strength dimensionally stable heavy weight hybrid reinforcing media at core and finished suitably on top with calibrated minerals for protecting and enhancing durability. = 1.1 sqm per sqm

Single component modified bituminous primer with mixing 1:1 ratio = 0.25 kg per sqm

Items & materials

B. Consumption of Materials For New Items Related to Cement concrete pavement chapter 6 of NH SOR 2019-20

Item No. B-1 White Topping

Unit- cum

Materials Requirement

Coarse Aggregate: The maximum size of coarse aggregate shall not exceed 31.5mm, 26.5 mm & 11.2 mm nominal size (conforming to clause 602.2.6.2 & 602.2.6.4 of MoRT&H's 5th revision 2013) graded stone chips @ 0.9 cum / cum. of Compacted vol.

Fine Aggregate : Coarse Sand (conforming to clause 602.2.6.3 & 602.2.6.4 of MoRT&H's 5th revision 2013): @ 0.45 cum / cum of Compacted vol.

Cement: As per mix design Ref : IRC : 44

Steel: 25 mm dia MS Dowel bars of Grade I conforming to IS: 432 @ 1.005 kg/cum. of Compacted vol.

Steel: 10 mm dia Deformed steel Tie bars of Grade Fe 500 conforming to IS: 1786 @0.990 kg / cum. of Compacted vol.

Joint Sealant = 0.74 Kg / cum. For construction joint only.

Sealant Primer =0.10 Kg / cum. of Compacted vol.

Plastic sheath 0.5mm thick for Dowel bars =0.044 sqm. / cum. of Compacted vol.

Curing Compound: 1.76 lit. / cum. of Compacted vol.

Super Plasticizer Admixture IS Marked as per 9103 - 1999 @ 0.5 % by wt. of Cement = 1.97 kg / cum of Compacted vol.

Fibres if incorporated in Mixed design to be considered in estimate separately.

Water: @ 205.71 lit. / cum. of Compacted vol.

Note : 1) Add 1% of material for cost of miscellaneous materials.

2) Quantities are only for estimation. For execution, grading shall conforming to table 600-3 of MoRT&H's 5th revision 2013.

3) Fly-ash upto 20% by wt. of cementitious material to be used in replace of OPC, such that OPC content shall not be less than 310 kg /cum. as per clause 602.3.2 of MoRT&H's 5th revision 2013.

Items & materials

Item No. B-3 Concrete Block Pavement.

Unit-sqm

Materials Requirement

Consumption of materials per sq.m.

Materials Requirement	Unit	Type-I 120 mm thickness	Type-II 100mm thickness	Type-III 80mm thickness	Type-IV 60mm thickness
(i)Paver Block (BIS marked)	sqm	1.05	1.05	1.05	1.05
(ii) Bedding Sand + Joint filling sand	cum	0.06	0.06	0.06	0.06

Items & materials

C. Consumption of Materials For New Items Related to Geosynthetics and reinforced earth Chapter 7 of NH SOR 2019-20

Item No. C-2 Geogrid for pavement reinforcement consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh having 50KN/sqm minimum tensile strength.

Unit-sqm

Geogrid for pavement reinforcement consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh. The geogrid should have minimum tensile strength upto 50KN/sqm in longitudinal and transverse direction having 25 x25 mm mesh opening. The material should comply with the specifications of **clause 708 of MoRT&H's 5th revision 2013.**

Materials Requirement

Glass fiber reinforced geogrid = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (Glass fiber geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.



Fiber Glass Geogrid (25 mm x 25 mm) mesh (FOR EXAMPLE)

Items & materials

Item No. C-3 Geogrid for pavement reinforcement consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh having 100KN/sqm minimum tensile strength.

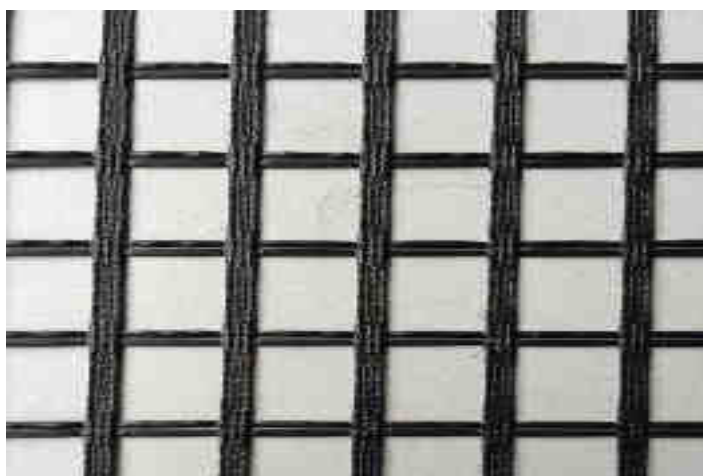
Unit-sqm

Geogrid for pavement reinforcement application consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh having upto 200kN/m minimum tensile strength in longitudinal and transverse direction, and having 12.5 x 12.5 mm mesh opening. The material should comply with specifications of **clause 708 of MoRT&H's 5th revision 2013.**

Materials Requirement

Glass Fiber reinforced Geogrid = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (Glass Fiber Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.



Fiber Glass Geogrid (12.5mm x 12.5mm) mesh (FOR EXAMPLE)

Items & materials

Item No. C-4 High modulus polypropylene bi-axial extruded geogrid for base/sub-base reinforcement in pavements of Minimum ultimate tensile strength of 15 KN/m (Ref to MoRT&H's 5th Revision 2013 Specification 708).

Unit-sqm



High modulus polypropylene bi-axial extruded geogrid for Item No. C-4, C-5 and C-6 (FOR EXAMPLE)

High modulus polypropylene bi-axial extruded geogrid conforming to Section 700 of MoRT&H's 5th revision 2013 specifications for base/sub-base reinforcement in pavements having

Minimum LCR/MIF value of 1.2 for subgrade $\text{CBR} \leq 3\%$,

Minimum ultimate tensile strength of 15 KN/m in both cross machine and machine direction, with Minimum 5KN/m & 7KN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction,

Junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.

Materials Requirement

High modulus polypropylene bi-axial Geogrids = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High modulus polypropylene bi-axial Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

Items & materials

Item No. C-5 High modulus polypropylene bi-axial extruded geogrid for base/sub-base reinforcement in pavements of Minimum ultimate tensile strength of 20 KN/m. (Ref to MoRT&H's 5th Revision 2013, Spec 708).

Unit-sqm

High modulus polypropylene bi-axial extruded geogrid confirming to Section 700 of MoRT&H's 5th revision 2013 specifications for base/sub-base reinforcement in pavements having

Minimum LCR/MIF value of 1.2 for subgrade CBR≤3%,

Minimum ultimate tensile strength of 20 KN/m in both cross machine and machine direction, with minimum 7kN/m & 14kN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction,

Junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.

Materials Requirement

High modulus polypropylene bi-axial Geogrids = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High modulus polypropylene bi-axial Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

Item No. C-6 High modulus polypropylene bi-axial extruded geogrid for base/sub-base reinforcement in pavements of Minimum ultimate tensile strength of 30 KN/m, (Ref to MoRT&H's 5th Revision 2013, Spec 708).

Unit-sqm

High modulus polypropylene bi-axial extruded geogrid confirming to Section 700 of MoRT&H's 5th revision 2013 specifications for base/sub-base reinforcement in pavements having minimum LCR/MIF value of 1.2 for subgrade CBR≤3%,

Minimum ultimate tensile strength of 30 kN/m in both cross machine and machine direction, with minimum 10kN/m & 21kN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction, Junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.

Materials Requirement

High modulus polypropylene bi-axial Geogrids = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High modulus polypropylene bi-axial Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

Items & materials

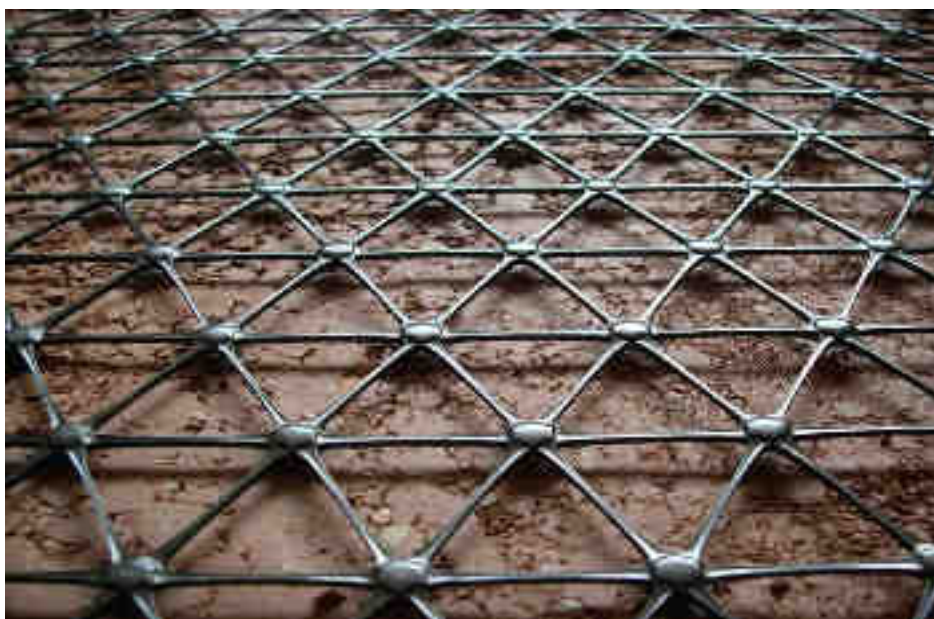
Item No. C-7. High radial stiffness and radial stiffness ratio polypropylene tri-axial punched and stretched geogrid with triangular aperture and hexagonal geometry confirming to ISO 10319/EOTA TR41 and with IRC accreditation for base/sub base stabilization in pavements with MIF value greater than 1.8 (IRC SP 59-2019-3.1.3 along with IRC : 37, 2018, clause no 7.2) for sub grade $CBR \leq 3\%$ with radial secant stiffness of 390 KN/m (tolerance 75 KN/m) at 5% strength , (MoRT&H's 5thRevision 2013,Table 700-7) and radial secant stiffness ratio of 0.7 to 0.8, junction efficiency greater than 90% with distance between alternate parallel ribs as 80mm.

Unit-sqm

Materials Requirement

Tri axial polypropylene geogrids as per approved design and specifications. = 1 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High modulus polypropylene Tri-axial geogrids) for accessories like tie-strips, nuts and bolts and loops/lugs for joining reinforcing elements with the facia pannels, overlaps and other protective elements for Tri-axial polypropylene geogrid and all other activities required to complete the item in all respect including taxes and transportation.

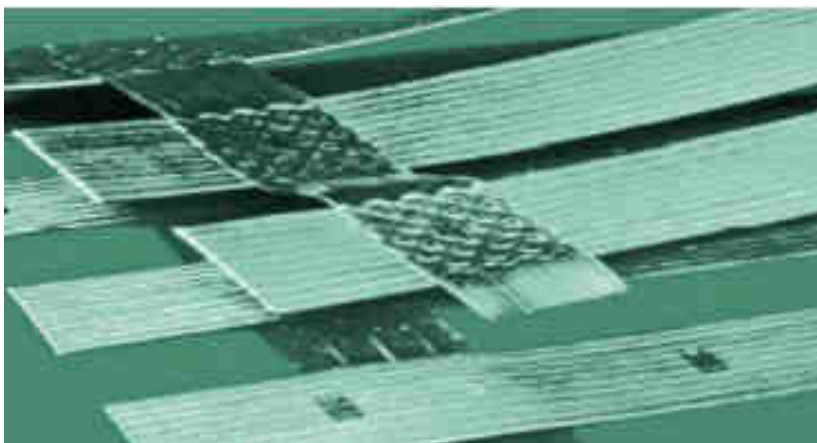


High modulus polypropylene Tri-axial geogrids(From Item No.7) (FOR EXAMPLE)

Items & materials

Item No. C-8 High strength flexible geogrids (HSFG) as soil reinforcement of different ultimate tensile strength.(Ref to MoRT&H's 5th Revision 2013, Spec 3100 & 703).

Unit-sqm



*High strength Flexible geogrid of various Ultimate Tensile strength (From It. No. 8(i) to 8 (iii)
(FOR EXAMPLE)*

Supplying and laying of High Strength Flexible geogrids (HSFG) as soil reinforcement / basal reinforcement as per MoRT&H's 5th Revision,2013,clause 3100 and IRC 113, made of high tenacity polyester core with polyethylene coating with Minimum Long Term Design Strength (LTDS) of more than 50% of ultimate tensile strength at 30 degree Celsius corresponding to 12 % strain.

i). High strength flexible geogrid of Ultimate Tensile strength of 500 kN/m

Materials Requirement

High strength flexible Geogrids = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High strength flexible Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

Items & materials

ii). High strength flexible geogrid of Ultimate Tensile strength of 900 kN/m

Materials Requirement

High strength flexible Geogrids = 1.0 sqm/sqm.

Add 10 per cent of the cost of reinforcing elements (High strength flexible Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

iii). High strength geogrid of Ultimate Tensile strength of 1200 kN/m

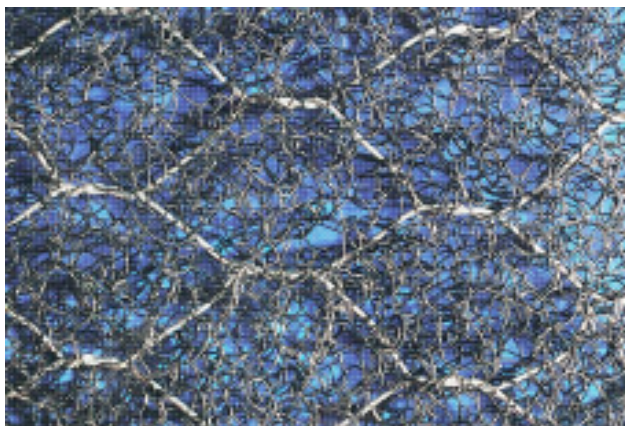
Materials Requirement

High strength flexible Geogrids = 1.0 sqm per sqm.

Add 10 per cent of the cost of reinforcing elements (High strength flexible Geogrid) for accessories like tie-strips, nuts and bolts and loops/lugs etc. for joining, lap length if required and transportation.

Item No. C- 9 Reinforced Geo composite made of Polypropylene Polymer made three dimensional matrix (Ref to MoRT&H's 5th Revision Specification 706, 2013).

Unit-sqm



Reinforced Geocomposite made of Polypropylene Polymer made three dimensional matrix of different mass per unit area. (From Item No.9 and 10 (FOR EXAMPLE))

Items & materials

Reinforced Geo composite made of Polypropylene Polymer made three dimensional matrix of minimum mass per unit area 450 grams/sq.m. extruded onto a mechanically woven double twisted hexagonal shaped steel wire Mesh and Mesh Type 6X8 having Zn coating.

The composite have a combined *mass per unit* and nominal thickness of 12mm; including fixing of U pins of 8mm diameter of depth 500mm at 1m c/c spacing.

- i) Double twisted hexagonal shaped steel wire Mesh having dia 2.2mm and the composite shall have a combined mass per unit area more than 1630 Grams / Sq.m.

Materials Requirement

3D geo mat including wastages due to overlaps = 1 sqm per sqm.

Steel for U pins - 8mm dia. And 500mm depth and spaced at 1m x 1m including wastage of 3% = 1 no per sq m.

- ii) Double twisted hexagonal shaped steel wire Mesh having dia 2.2/3.2mm and the composite shall have a combined mass per unit area more than 1920 Grams / Sq.m.

Materials Requirement

3D geomat including wastages due to overlaps = 1 sqm per sqm.

Steel for U pins - 8mm dia. And 500mm depth and spaced at 1m x 1m including wastage of 3% = 1 no per sq m.

Item No. C-10 Reinforced Geo composite made of Polypropylene Polymer made three dimensional matrix (Ref to MoRT&H's 5th Revision Spec 706, 2013).

Unit-sqm

Supplying and laying of Reinforced Geo composite made of Polypropylene Polymer made three dimensional matrix of minimum mass per unit area 450 grams/sqm extruded onto a mechanically woven double twisted hexagonal shaped steel wire Mesh and Mesh Type 8X10 having Zn coating .

The composite shall have a combined mass per unit area and nominal thickness of 12mm; including fixing of U pins of 8mm diameter of depth 500mm at 1m c/c spacing, at easily accessible location including top and bottom, with all leads and lifts,

Items & materials

- i) Double twisted hexagonal shaped steel wire Mesh having dia 2.7mm and the composite shall have a combined mass per unit area more than 1850 Grams / Sq.m.

Materials Requirement

3D geomat including wastages due to overlaps = 1 sqm per sqm.

Steel for U pins - 8mm dia. And 500mm depth and spaced at 1m x 1m including wastage of 3% = 1 no per sq.m.

- ii) Double twisted hexagonal shaped steel wire Mesh having dia 2.7/3.7mm and the composite shall have a combined mass per unit area more than 2130 Grams / Sqm.

Materials Requirement

3D geomat including wastages due to overlaps = 1 sqm. per sqm.

Steel for U pins - 8mm dia. And 500mm depth and spaced at 1m x 1m including wastage of 3% = 1 no per sq m.

Item No.C-11. Woven Geotextile as per Clause No. 702 on the prepared subgrade as the separator cum reinforcement with necessary overlap as per drawing. (Ref to MoRT&H's 5th Revision Spec 700, 2500, 2013).

Unit-sqm



Woven Geotextile of Item No C-11(FOR EXAMPLE)

Items & materials

Materials Requirement

i) For Type-I (160gsm) woven Geotextile

Woven Geotextile as per approved design and specifications. = 1 sq.m. per sq.m.

ii) For Type-II (240gsm) woven Geotextile

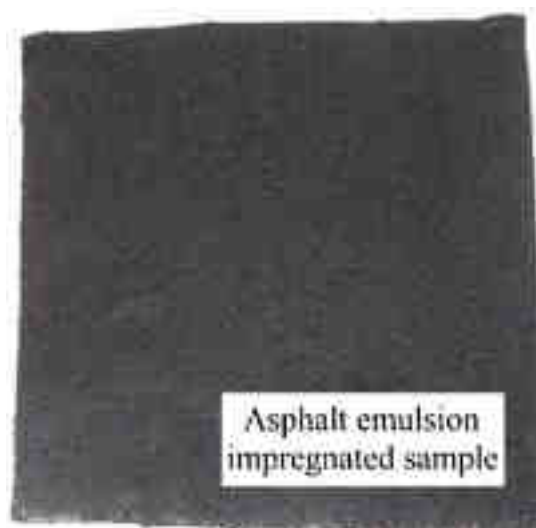
Woven Geotextile as per approved design and specifications. = 1 sq.m. per sq.m.

iii) For Type-III (450gsm) woven Geotextile

Woven Geotextile as per approved design and specifications. = 1 sq.m. per sq.m.

Item No. C-12 Providing and laying a fibre glass geogrid manufactured from high quality glass fibre strands and coated with a polymer modified bitumen as reinforcement to Asphalt overlay flexible pavements and Asphalt over distressed rigid P.C.C (Ref to MoRT&H's 5th revision 2013 Spec 708).

Unit-sqm



Fiber glass geogrid manufactured from high quality glass fiber strands and coated with a polymer modified bitumen (From Item No.12 (i) and (ii))

(FOR EXAMPLE)

Items & materials

Materials Requirement

i) **For Tensile strength-50 kN/m**

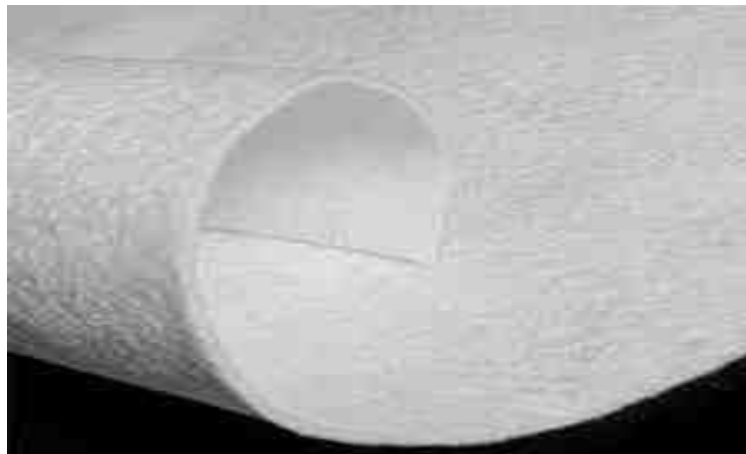
Fiber Glass Geogrids as per approved design and specifications = 1 sqm per sqm.

ii) **For Tensile strength-100 kN/m**

Fiber Glass Geogrids as per approved design and specifications = 1 sqm per sqm.

Item No. C-13 Providing and laying Needle Punched and mechanically bonded non-woven Geotextile indigenously manufactured from high quality fibers on the prepared subgrade for Separation.(Ref to MoRT&H's 5th Revision 2013, Spec- 702).

Unit-sq.m.



Punched and mechanically bonded non-woven Geotextile for the Item No C-13

(FOR EXAMPLE)

Items & materials

Materials Requirement

i) **For Type-I (150gsm) non woven Geotextile**

Non -woven Geotextile as per approved design and specifications. = 1 sq.m per sq.m.

ii) **For Type-II (200gsm) non woven Geotextile**

Non-woven Geotextile as per approved design and specifications. = 1 sq.m per sq.m.

iii) **For Type-III (450gsm) non-woven Geotextile**

Non-woven Geotextile as per approved design and specifications. = 1 sq.m per sq.m.

Item No.C-14 One filtering UV stabilized polypropylene nonwoven geotextile. (Ref to CPWD SOR 2016, Sub Head 26 for New Technologies and Materials Code No 8960).

Unit-sqm



One filtering UV stabilized polypropylene nonwoven geotextile. (Item No. 14 & 15)

(FOR EXAMPLE)

Drainage composite for use behind walls having three dimensional composite with thermo bonding a draining core having “W” configuration in extruded monofilaments of minimum thickness 6.9 mm with one filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75mm.

Tensile strength of 8.0 KN/m that will be working as separation or protecting layer, geo composite having in plane flow capacity of 2.60 L / (m.s) at hydraulic gradient of 1.0 at 20 k pa pressure and tensile strength of 12 KN/m.

Mass per unit area of 540 gsm supplied in the form of roll for easy transportation to site of work as per detailed specification all complete as per directions of Engineer in charge.

Items & materials

Materials Requirement

Polypropylene nonwoven geotextile drainage Composite as per approved design and specifications = 1 sqm per sqm .

Add 10 per cent of the cost of Polypropylene nonwoven geotextile drainage composites for accessories for joining seats with the fascia panels, overlaps and other protective elements for Polypropylene nonwoven geotextile Drainage composites and other miscellaneous activities required to complete the item in all respect including transportation and takes.

Note: Lower gsm may be used in RE structure.

Item No. C-15. Two filtering UV stabilized polypropylene nonwoven geotextile. (Ref to CPWD SOR 2016, Sub Head 26 for New Technologies and Materials Code No 8960).

Unit-sqm

Drainage composite for use behind walls, between two different fills, alongside drains of road, below concrete lining of canals etc. having three dimensional composite with thermo bonding a draining core having “W” configuration in extruded monofilaments of minimum thickness 7.2mm, with two filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75mm.

Tensile strength of 8.0 KN/m that will be working as separation or protecting layer, geo composite having in plane flow capacity of 2.3 L / (m.s) at hydraulic gradient of 1.0 & 20 kPa pressure.

Tensile strength of 18 KN/m , with mass per unit area of 740 gsm, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials, labour etc. complete and as directed by Engineer - In - Charge.

Materials Requirement

Polypropylene nonwoven geotextile Composite Drainage as per approved design and specifications = 1 sqm per sqm.

Add 10 per cent of the cost of Polypropylene nonwoven geotextile drainage Composite for accessories for joining seats with the fascia panels, overlaps and other protective elements for Polypropylene nonwoven geotextile drainage composites and other miscellaneous activities required to complete the item in all respect including transportation and takes.

Note: Higher gsm may be used in drainage composite at the time of pavement construction.

Items & materials

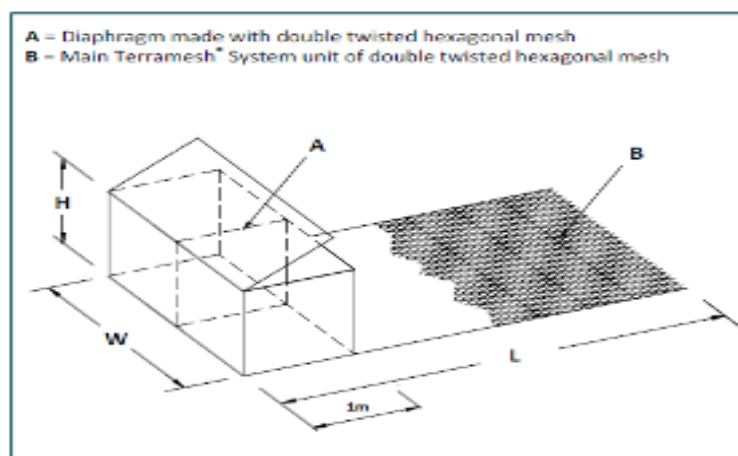
Item No C-16. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRTH's 5th revision 2013, clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedge with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm. = 1 cum per cum



BIS certified Gabion fascia box of different height (Item No. C-16 to C- 24)

(FOR EXAMPLE)

Items & materials

Item No. C-17. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit)with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012,MoRTH's 5threvision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedge with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm = 1 cum per cum.

Item No. C-18. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit)with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012,MoRTH's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm = 1 cum per cum.

Items & materials

Item No. C-19 Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Item No. C-20. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Items & materials

Item No. C-21. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Item No. C-22. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012,, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm=1 cum per cum.

Items & materials

Item No. C-23 Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1cum per cum.

Item No. C-24. Providing & making BIS certified Gabion fascia box.

Unit-cum

Providing & making BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy + Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Items & materials

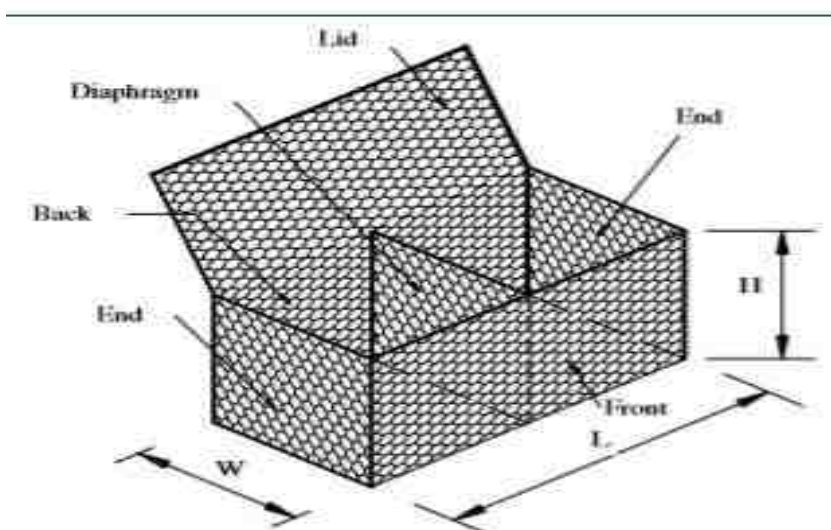
Item No. C-25 Providing & making BIS certified Gabion fascia box

Unit-cum

Providing & making Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500 of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.



BIS certified Gabion fascia box of different height (Item No.C-25 to C-33)

(FOR EXAMPLE)

Items & materials

Item No. C-26. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500 of required size, Mesh Type 10x12(D=100 mm with tolerance of $\pm 2\%$), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1.0 cum per cum.

Item No. C-27. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x0.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of $\pm 2\%$), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Items & materials

Item No. C-28. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of $\pm 2\%$), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Item No. C-29. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of $\pm 2\%$), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm=1.0 cum per cum.

Items & materials

Item No. C-30. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x0.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1cum per cum.

Item No. C-31. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum

Providing & making Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1 cum per cum.

Item No. C-32. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making Gabion structure of size (2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with

Items & materials

partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 200mm= 1.0 cum per cum.

Item No. C-33.Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500

Unit-cum

Providing & making Gabion structure of size (2x0.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of $\pm 2\%$), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in- charge.

Materials Requirement

Stone boulder with least dimension 200mm=1 cum per cum.

Item No. C-34.Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

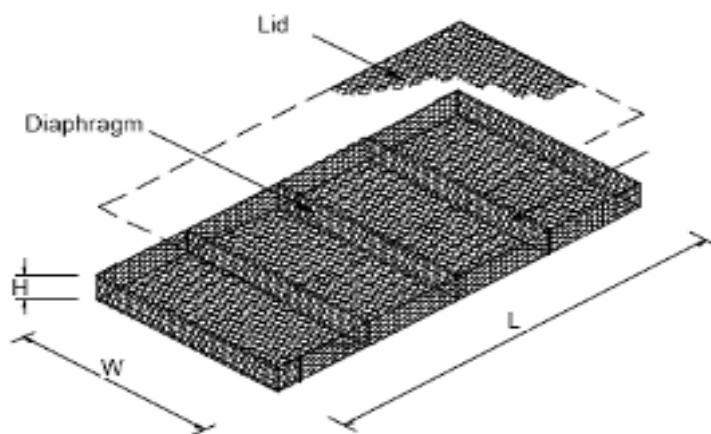
Unit-cum

Providing & making BIS certified Gabion structure of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Items & materials

Materials Requirement

Stone boulder with least dimension 200mm=1 cum per cum.



BIS certified Gabion fascia box of different height (Item No. C-34 to C-36)

(FOR EXAMPLE)

Item No. C-35. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.23x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy + Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement.

Stone boulder with least dimension 90-120mm= 1 cum per cum

Items & materials

Item No. C-36. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy + Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1 cum per cum.

Item No. C-37. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

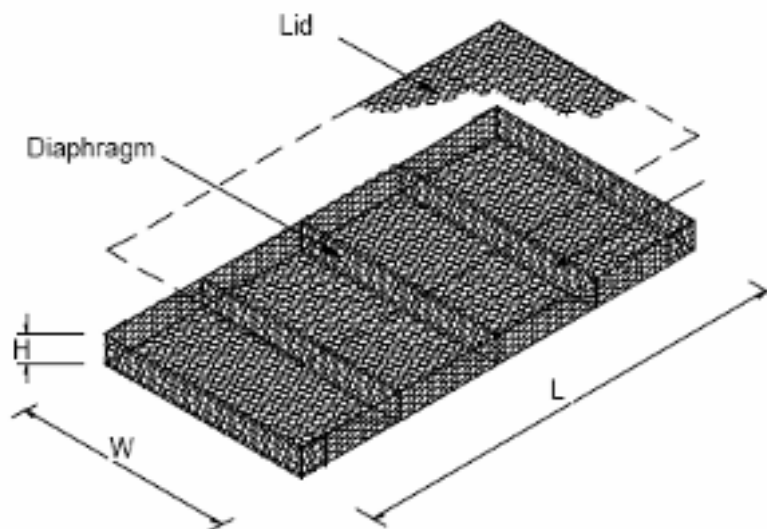
Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn + Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1 cum per box cum.

Items & materials



BIS certified Gabion fascia box of different height (Item No. C-37 to C-39)

(FOR EXAMPLE)

Item No. C-38. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.23x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1 cum per cum.

Items & materials

Item No. C-39. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1cum per cum.

Item No. C-40. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

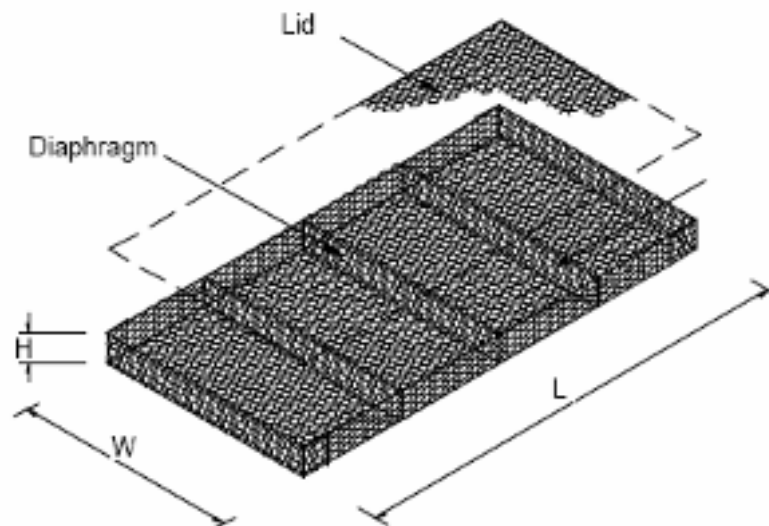
Unit-cum.

Providing & making BIS certified Gabion structure w of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500. of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh wire diameter 2.20 mm, mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1 cum per cum.

Items & materials



BIS certified Gabion fascia box of different height (Item No. C-40 to C-42)

(FOR EXAMPLE)

Item No. C-41. Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.23x1)m e with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh wire diameter 2.20 mm, mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm=1 cum per cum.

Items & materials

Item No. C-42 Gabion structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500.

Unit-cum.

Providing & making BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5th revision 2013 clause 2500. of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh. wire diameter 2.20 mm, mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.

Materials Requirement

Stone boulder with least dimension 90-120mm =1 cum per cum.

Item No C-43.Cylindrical Gabions /Sack Gabions structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh.

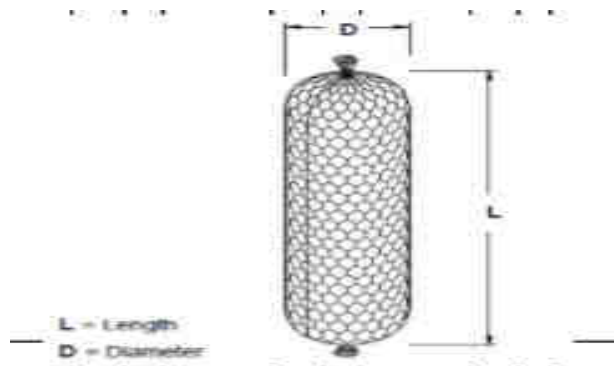
Unit-cum.

Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn + 10% Al alloy +Polymer Coated wire mesh, Mesh Size 10x12. The approximate minimum weight of the sack gabion would be 1350 kg when filled as per detailed technical specification.

Materials Requirement

Quantity of sand required = 1 cum. Per cum.

Items & materials



Cylindrical Gabions /Sack Gabions structure with Mechanically Woven Double Twisted Hexagonal Shaped of different coated Wire mesh (Item No. C-43to C-45) (FOR EXAMPLE)

Item No. C-44.Cylindrical Gabions /Sack Gabions structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh.

Unit - cum.

Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn +Polymer Coated wire mesh, Mesh Size10x12. The approximate minimum weight of the sack gabion would be 1350 kg when filled as per detailed technical specification.

Materials Requirement

Quantity of sand required = 1 cum. Per cum.

Item No. C-45.Cylindrical Gabions /Sack Gabions structure with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh.

Unit - cum.

Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn Coated wire mesh, Mesh Size10x12. The approximate minimum weight of the sack gabion would be 1350 kg when filled as per detailed technical specification.

Materials Requirement

Quantity of sand required = 1 cum. Per cum.

Note: - The photography are provided to give an idea about the product associated with the particular item. It is necessarily not meant to use the product of a particular manufacturer but the specification has mentioned in the item should be strictly followed.

Items & materials

D)Consumption of Materials For New Items Related to Traffic signs, Marking and other road appurtenances chapter 8 of NH SOR 2019-20

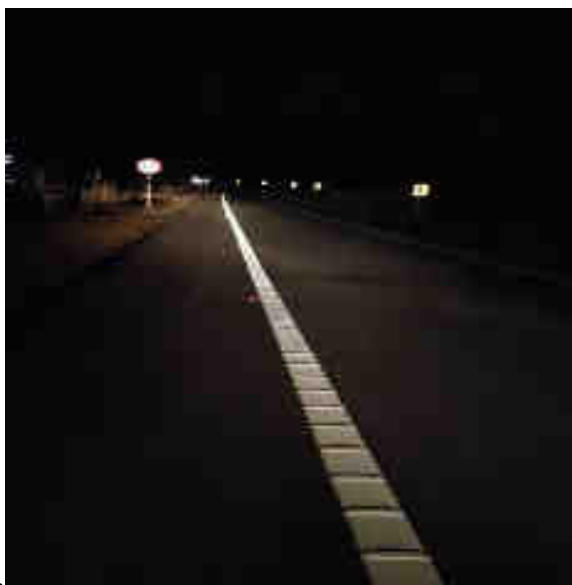
Item No. D-1Raised Profile Edge Markings with Reflectorized Hot Applied Thermoplastic Compound on Bituminous Road surfaces.

Unit -sq. m.

Raised Profile Edge Markings with Hot Applied Thermoplastic Road Markings along with Drop on Glass Beads as per the Provisions laid in IRC 35-2015.

The height of the rib should be 8 mm for normal roads and it may extend upto 11 mm in case of expressway.

The general gap between each rib shall vary between 200-500 mm, as per the direction of the Engineer in charge.



Raised Profile Edge Markings with Reflectorized Hot Applied Thermoplastic Compound for Item no. D-1
(FOR EXAMPLE)

Items & materials

Specifications for Raised Profile Edge Line Material

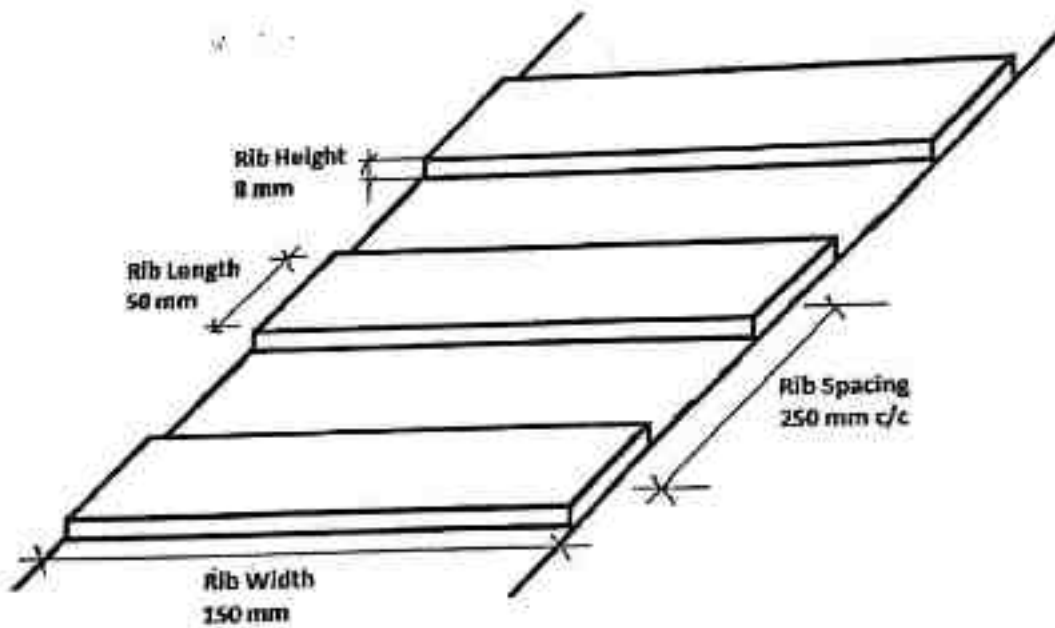
Following are the specifications of the material to be adopted for the provision of Rib Line Edge Marking as per the Australian Standard AS 4049.2-2005

Sl No.	Property	Value
1	Colour	White: Equivalent to or whiter than Y35 off white (AS2700S). Yellow: Equivalent to or whiter than Y14 golden yellow (AS2700S).
2	Luminance	Not less than 75%
3	Softening Point	100-110 degree Celsius (AS 2314.18)
4	Initial Retro reflectivity	Minimum 250 mod/lux/m2
5	Retro reflectivity after one month	Minimum 150 mod/lux/m2
6	Skid Resistance	Minimum 45 BPN
7	Binder	18-20% by weight
8	Glass Beads	20% by weight
9	Mineral Content	50-60% by weight (including aggregate, pigment & extender)

ANNEXURE-II

Drawing for the Raised Profile Edge Line

Adopt the following Drawing for the provision of Raised Profile Edge Line:



Drawing: Rib Line Edge Marking



Scanned with
CamScanner

Note: The rate of ribbed edge line marker is as per Ministry's Circular Ref. No. RW/NH/12013/11/6/SP (B - 6)/EAP, Dated-3rd January 2019.

Items & materials

Item No. D-2. Hybrid Solar-Retro Reflective Chevron Signs.

Unit - each

Solar Powered Retro Reflective LED Blinking Chevron Signs as per the following specifications Solar panel: 450 MaH/9v/4 watts, Battery: Li-ion 6 Volt 4.5 AH.C-10, Back up time: 108 hrs, Led: 112-130 nos.

Display thickness: ACP or Aluminum 2 mm thick, Powder coated body. The Unit shall be self-sufficient and shall be mounted on a Circular Pole: 50 NB (medium) duly painted and grouted into the soil using M15 concrete (45 cm x 45 cm x 60 cm) Size: 600 mm x 450 mm. The Retro reflective sheeting shall be HIP, Type IV as per IRC 67.



Hybrid Solar-Retro Reflective Chevron Signs for D-2(FOR EXAMPLE)

Item No. D-3. Solar Powered Road Studs: -

Unit - each.

Solar powered road studs rectangular/circular shape (with super bright LEDs for long visibility from distance greater than 800m) having twin shank and body made of aluminum alloy or polycarbonate material which shall be absolutely weather resistant.

Strong enough to support a load of 13635 kg tested in accordance with **ASTM-D 4280** and water resistance as per IP65 in reference with **IS 12063:1987**, fixed to the road surface using adhesives and the procedure recommended by manufacturer as per clause no **804.6** and **804.7** of MoRT&H's 5th Revision 2013.

Items & materials



Solar Powered Road Studs for Item No. D-3(FOR EXAMPLE)

Item No. D-4 Wire Rope Safety Crash Barrier

Unit - Running Metre.

Wire Rope Safety Barrier, meeting and conforming to **IRC:119 :2015**, Annexure II, Sec 4. The barrier shall be supplied complete with steel posts (spacing 3.0m c/c), Post length 1200mm, Post Cross Section not less than 100 x 50 x 16 mm, thickness 4 mm, plastic cap for post with retro-reflective sheet, plastic cable spacers, Stainless Steel Strap, rectangular steel sections having thickness 3mm and length 420mm to be used as post foundation sleeves, plastic top covers and bottom caps for steel sleeves, 3 galvanized wire ropes running along the entire length, each wire rope 19mm diameter, (3x7)-pre-stretched-16.7 Ton capacity, required steel fittings, steel turnbuckles, steel anchors and M30 concrete complete.

All steel components of the barrier shall be hot dip galvanized, having an average zinc coating thickness not less than 75 microns.

The installation shall include excavating in soft soil for post foundations (250 mm dia x 610 mm depth at 3m c/c), excavation in all kind of soil for end anchor foundations (1m x 1m x1.5m each, 2 for each section), grouting of steel sleeves up to depth of 420 mm in each post foundation with M30 concrete, grouting anchor plates in each end anchor foundation with

Items & materials

M30 concrete, placing of posts in the grouted steel sleeves ensuring that the posts and barrier are in line and level.

Note: The bidder shall enclose all relevant papers in support of meeting the above criteria, the bidder shall also enclose Authorization from the OEM.



Wire Rope Safety Crash Barrier Item No D-4(FOR EXAMPLE)

Item No. D-5 Reflective Pavement Marker with Shank

Unit - each.

Single mould twin shanked moulded road markers/ road studs made of poly carbonate or ABS body and shall support a load of 13635 kg tested in accordance to ASTM D4280.

Complying with specifications as per clause 804 of MoRT&H 5th revision 2013 and section 5 of IRC-35:2015, the reflecting panels shall consist of number of lenses containing single or dual prismatic cubes capable of providing total internal reflection of the light entering the lenses face.

The slope of retro reflecting surface should be 25 +/- 5 degree to base. The reflective marker shall be fixed on the road surface using the epoxy/adhesive and the procedure recommended by the manufacturer. No nails shall be used to affix the markers so that they do not pose safety hazard on the road.

Items & materials



Reflective Pavement Marker with Shank for D-5(FOR EXAMPLE)

Item No. D-6 Median Marker:

Unit - each.

Made up of tough, impact resistant thermoplastic body, reflectorized on both side using apx. 85 x 85 mm (for square shape or equivalent in case of round shape Fluorescent yellow prismatic sheeting (type-XI) reflective surface should be incline to approximate 20 degree to vertical to provide maximum reflective. Height & width of median marker shall be approximate by 110 mm complete with all fixing arrangements.



Median Marker for Item No D-6(FOR EXAMPLE)

Items & materials

Item No. D-7 Retro-reflective studs

Unit - each.

Metal Road studs with shank, made of Die-cast Aluminium alloy moulded body with the size 100x100x19.8 mm, shank size 50 mm, having reflective panels of 34 glass reflective elements embedded in plastic and 19 degrees tilted to provide the highest reflectivity having reflective panels in yellow, red and white plastic with yellow, red, white or green glass elements. The weight of Aluminium stud should be not less than 370 grams.

The stud shall have a compressive strength test certificate of 40 tons in accordance with **ASTM D 4280-04, clause no 806 of 5 th revision of MoRT&H's of 2013** and successfully passing retro reflectivity test as per EN 1463 part I & part II, from a reputed international laboratory.

The fixing of road studs shall be done by using a recommended Two Component Epoxy after drilling 60 mm hole on the road for the shank to go inside.

Materials Requirement

Die - cast Aluminium Alloy Road Stud having reflective glass element - 1 no each.



Retro-reflective studs for D-7(FOR EXAMPLE)

Items & materials

Item No. D-8 Rumble marking using two component cold plastic

Unit - Sqm.

Rumble strips with two component cold plastic profiled material to give total width of 500 mm and height of 10 mm.

Ramp provided to give width of 145 mm, width of ribs to be 55mm laid at center to center distance of 35 mm, with reflective glass beads @ 350 gm / sqm.

The finished surface to be levelled, uniform and free from streaks and holes, to be applied on the edge lines, including secondary surface cleaning and cost of all materials etc. complete.

Materials Requirement

Two Component Cold Plastic = 12 kg per sqm.

Hardener = 0.005 kg per sqm.

Thinner = 0.08 litre per sqm.



Rumble marking using two component cold plastic Item No D-8

(FOR EXAMPLE)

Items & materials

Item No. D-9 Audible Vibratory Profile Marking with Hot Applied Thermoplastic and Reflectorizing Glass.

Unit - Square Metre.

Raised profile edge line marking (audible vibratory) with special thermoplastic road marking compound 2 mm thick base coat layer above that ribs profile of 6 mm thick (Total 8 mm thick) at the distance of 250 mm between two ribs including reflectorizing glass beads @ 250 gm. / sq. m. area.

The minimum and maximum width of raised profile should be 100 mm and 150 mm. The thickness of 8 mm profile should be exclusive of surface applied glass beads. The finished surface to be levelled, uniform and free from streaks and holes, to be applied on the edge lines, including secondary surface cleaning and cost of all materials etc. complete. According to IRC: 35-2015, Clause 7.7, Section 800, clause 803.7 of MOST 2013.

Materials Requirement

Hot applied thermoplastic compound = 10 kg per sq.m.

Reflectorizing glass beads = 300 gm. per sq.m.



Audible Vibratory Profile Marking with Hot Applied Thermoplastic and Reflectorizing Glass for D-9

(FOR EXAMPLE)

Items & materials

E) Consumption of Materials for New Items Related to Bridge Structure of NH SOR 2019-20

Item No. E-1. Per MT of Supplying fitting and placing mild steel plate in bridge work (Super structure - Supplying of all types of structural steel)

Materials Requirement

Plate Girder/ Semi through Girder

Mild steel plates (including wastage 5%) = 1.05 MT per MT

Rivets (including riveted connections @3%) = 0.0315 MT per MT

20mm dia holding down bolts = 0.000657 MT per MT

Item No. E-2per MT of Supplying fitting and placing mild steel plate in bridge work (Super structure - Supplying of all types of structural steel)

Materials Requirement

Open Web Girder

Mild steel plates (including wastage 5%) = 1.05 MT per MT

Rivets (including riveted connections @3%) = 0.0315 MT per MT

20mm dia holding down bolts = 0.000127 MT per MT

Item No. E-3. Metalizing of steel work of girder with grit blasting per sq. m.

Materials Requirement

One coat of primer (IS 5666) = 0.054 litre per sq. m.

One coat of Zinc chrome primer (IS 104) = 0.054 litre per sq. m.

Two coat of Aluminium paint as per IS 2339 = 0.090 litre per sq. m.

Items & materials

Item No. E-4. Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required.

Materials Requirement

With one coat of ready mix red lead conforming to IS 102

a) Painting Work of Bridge Girder upto 30.5 m.

Red oxide Zinc chromate primer = 0.054 litre per sq. m.

b) Painting Work of Bridge Girder above 30.5 m

Red oxide Zinc chromate primer = 0.054 litre per sq. m.

Item No. E-5. Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required.

With one coat of ready mix Zinc Chromate conforming to IS 104 followed by one coat of Zinc Chromate red oxide conforming to IS: 2074

Materials Requirement

a) Painting Work of Bridge Girder up to 30.5 m.

Red oxide primer in two coats

Red oxide Zinc chromate primer (for first coat) = 0.054 litre per sq. m.

Red oxide Zinc chromate primer (for second coat) = 0.036 litre per sq. m.

b) Painting Work of Bridge Girder above 30.5 m

Red oxide Zinc chromate primer (for first coat) = 0.054 litre per sq. m.

Red oxide Zinc chromate primer (for second coat) = 0.036 litre per sq. m.

Item No. E-6 Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required.

With two coats Aluminium paints in dual containers conforming to IS 2339

Materials Requirement

a) Painting Work of Bridge Girder upto 30.5 m

Aluminium Paint (for first coat) = 0.054 litre per sq. m.

Aluminium Paint(for second coat)= 0.036 litre per sq. m.

Items & materials

b) Painting Work of Bridge Girder above 30.5 m

Aluminium Paint (for first coat) = 0.054 litre per sq. m.

Aluminium Paint (for second coat) = 0.036 litre per sq. m.

Item No E-7 Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required.

With two coats epoxy paint

a) Painting Work of Bridge Girder upto 30.5 m

Epoxy Paint (for first coat) = 0.054 litre per sq. m.

Epoxy Paint(for second coat)= 0.036 litre per sq. m.

b) Painting Work of Bridge Girder above 30.5 m

Epoxy Paint (for first coat) = 0.054 litre per sq. m.

Epoxy Paint(for second coat)= 0.036 litre per sq. m.

CHAPTER - 17

MISCELLANEOUS

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
A)	<u>New items related to bases and surface courses (Bituminous) Chapter 5 of NH SOR 19-20</u>			
A-1	Mechanised Mastic Asphalt			
	Providing and laying mastic asphalt by mechanical means done in a centralised batching plant with appropriate built- in pollution control units with specified aggregates and lime stone powder (with Calcium Carbonate content not less than 80%) and approved grade of industrial grade bitumen as per approved design mix, transporting the hot mastic asphalt to the worksite by thermocontainer with the arrangement of maintaining specified temperature during transport, laying with a paver finisher for spreading and floating the plastic mix uniformly on the surface to the specified width and thickness at correct grade and camber through its heated, distortion free steel float and turning sword distributors, applying uniform size of bitumen coated stonechips at a specified rate, on the laid surface through a power driven chip spreader as per the procedure laid down in Annexure - II of IRC : 107 (2013) all complete as per clause 516 along with IRC : 107 for materials and design mix etc.	sqm	200.00	Material cost to be added. Consumption of materials to be followed as per chapter - A Sl. No. A-1.
A-2	Waste Plastic Modified Dense Graded Hot Bituminous mix for wearing coarse			
	Providing and laying waste plastic modified dense graded bituminous mix for wearing coarse of a flexible pavement with HMP of capacity not less than 60-90TPH with approved aggregate and specified grade of bitumen, the bituminous mix be suitably modified with waste plastic as per table 2 of IRC : SP:98 shredded to a specified size, making the waste plastic free from impurities, as per " Dry Process" , compacted to specified grade, level, camber and thickness to a specified density including collection of waste plastic, cleaning and shredding by shredding machine all complete as per IRC :SP: 98.	cum	1561.60	Material cost to be added. Consumption of materials to be followed as per chapter - A Sl. No. A-2.
A-3	Water proofing work with the provision of Asphalt Seal pad for bridge deck, Parking Decks , Bridge Ramps , Elevated Roadways, Pedestrin Subways (concrete surface).			
	Providing and Laying 5mm thick Asphalt seal pad made of actatic polypropylene polymer APP modified bituminous pad reinforced with non woven polymer mat of high bear strength as a carrier confirming to ASTM - D6222 (S) specification along with the single component with antiskid material grafted on the asphalt seal pad over concrete surface bridge deck slab complete as per specifications.	sqm	367.31	Complete Rate. See the detailing of the item on chapter-A, Sl no. A- 3

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
B)	<u>New items related to cement concrete pavement Chapter 6 of NH SOR 19-20</u>			
B-1	White Topping			
	Providing and laying conventional or thin white topping as per IRC:SP 76 over bituminous surface(existing bituminous surface preparation to be done if neccesary either by milling and cleanning or levelling course with bituminous mix or concrete mix as directed by E-I-C or as shown in drawing to correct surface profile (and to be paid seperately) or short panel concrete pavement over prepared base with pavement quality concrete (PQC) mixed in batching plant with weigh batching facility as per approved mix design as per IRC: 44 to achieve minimum flextural strength of concrete of 4.50 Mpa at 28 days, transported to site by transit mixer, laid with sensor paver finisher,spread, vibrated, compacted and finished in continuous operation to the lines and grades with contraction joint and longitudinal joint (including dowel bars and tie bars as per clause no 602.2.8) as per drawing, texturing / tinning (as per clause no 602.9.11 of MoRT&Hss 5th revision) of surface, curing as per(clause no 602.9.12), saw cutting to a depth of 1/3 to 1/4 of PQC slab, width 3 mm - 4 mm and grid as per design and drawing within the specified time (saw cutting to be paid seperately in RM) including cost of formwork, labour, plant and machinerics etc complete as per specification.	cum	1645.20	<i>For consumption of materials refer Chapter A, Sl. No. B-1. Only cost of stone Aggeragate, Sand, Cement and Steel to be added. Fibres if incorporated in Mixed design to be considered in estimate seperately (Dose - As per manufacturer specification). Cost of super plasticizer is included. For longitudinal joint (if provided, in case of more than single lane) and contraction joint,provision of Tie bar and dowel bar has to be made respectively. For Tie bar 10mm dia HYSD bar @750 mm c/c for and for Dowel bar 25 mm dia MS bar @ 300 mm c/c has to be provided in estimate seperately.</i>
B-2	Concrete Sawing			
	Concrete Sawing of the PQC slab upto the specified depth and width as per drawing and design, done strictly within the specified time as mentioned in the drawing or as per direction of E- in -C .by concrete sawing machine operated either by electric or by petrol/diesel fitted with water tank with flexible hoss and pump and diamond edge sawing blades.	rm		<i>Complete rate</i>
	a)For depth of cutting upto 100 mm th.		18.00	
	b) For depth of cutting above 100 mm th and upto 150 mm th.		27.00	

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
B-3	Concrete Block Pavement			
	Supplying and Laying in the particular pattern of cement concrete block pavements with BIS certification and spacer nib of size and shape with aspect ratio as per IS 15658 with cement concrete grade not less than M40 on the prepared base course of specified CBR as mentioned in design and drawing with a cushion of bedding sand of 30 to 50 mm thick (grading of the bedding sand as per clause 6.5.1 of IRC SP-63) and filling up the gaps between two paver block with joint filling sand (the grading of the joint filling sand will be as per clause 6.5.2) complete as per IRC SP - 63 with proper confinement of bedding and joint filling sand, compaction,levelling and filling up of the edge gaps with mortar. Type I, II,III, IV should be as per design requeriment or instruction of E-I-C.			<i>Cost of bedding and jointing sand is not included in the item. It will be paid seperately as per chapter A, Item no. B-3. Edge protection should be treated as separate item.</i>
	Note : The IRC SP 63 will prevail out of the two codes			
	a) Type - I (120 mm thickness) inter locking system	sqm	151.00	
	b) Type - II (100 mm thickness) inter locking system	sqm	145.00	
	c) Type - III (80 mm thickness) inter locking system	sqm	139.00	
	d) Type - IV (60 mm thickness) inter locking system	sqm	132.00	

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C)	<u>New items related to geosynthetics and reinforced earth Chapter 7 of NH SOR 19-20</u>			
C-1	Installation of Pre fabricated Vertical Drain with drainage composite wrapped in sand layer			
	Installation of PVD of 100mm width with rod by the help of mechanical rig. The pre fabricated vertical drain material shall be cut nearly at the upper end with 100 mm to 200 mm protruding above the working surface minimum 100 mm sand blanket is provided wrapped in a layer of drainage composite in accordance with clause 7, of IS - 15284 (Part - II, 2004) Prefabricated Vertical Drain (PVD) UV protected formed of Polypropylene of core width as per MoRT&H's section - 704, Table - 700-11 and 314 of MoRT&Hs specification (5th Revision)	rm	150.00	<i>Rate of sand bed with medium sand to be paid seperately</i>
C-2	Supplying and laying geogrid for pavement reinforcement application consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh having upto 50kN/m minimum tensile strength in longitudinal and transverse direction having 25 X 25mm mesh opening, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials, labour etc. complete and as directed by Engineer - In - Charge. (Complying with specifications of clause 708 of MoRT&Hs 5th Revision.)	sqm	79.50	<i>Complete rate See the detailing of the item on chapter-A, Sl no. C-2</i>
C-3	Supplying and laying geogrid for pavement reinforcement application consisting of glass fiber strands arranged in a grid shape with squared polymeric coated mesh having upto 200kN/m minimum tensile strength in longitudinal and transverse direction, and having 12.5 x 12.5 mm mesh opening, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials, labour etc. complete and as directed by Engineer - In - Charge. (Complying with specifications of clause 708 of MoRT&H's 5th Revision.)	sqm	109.20	<i>Complete rate See the detailing of the item on chapter-A, Sl no. C-3</i>
C-4	Supplying and laying of high modulus polypropylene bi-axial extruded geogrid conforming to Section 700 of MoRT&Hs 5th Revision specifications for base/sub-base reinforcement in pavements having minimum LCR/MIF value of 1.2 for subgrade CBR≤3% %, minimum ultimate tensile strength of 15 kN/m in both cross machine and machine direction, with minimum 5kN/m & 7kN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction, junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.	sqm	71.80	<i>Complete rate See the detailing of the item on chapter-A, Sl no. C-4</i>

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-5	Supplying and laying of high modulus polypropylene bi-axial extruded geogrid confirming to Section 700 of MoRT&Hs 5th Revision specifications for base/sub-base reinforcement in pavements having minimum LCR/MIF value of 1.2 for subgrade CBR≤3%, minimum ultimate tensile strength of 20 kN/m in both cross machine and machine direction, with minimum 7kN/m & 14kN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction, junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.	sqm	71.80	Complete rate See the detailing of the item on chapter-A, Sl no. C-5
C-6	Supplying and laying of high modulus polypropylene bi-axial extruded geogrid confirming to Section 700 of MoRT&Hs specifications for base/sub-base reinforcement in pavements having minimum LCR/MIF value of 1.2 for subgrade CBR≤3%, minimum ultimate tensile strength of 30 kN/m in both cross machine and machine direction, with minimum 10kN/m & 21kN/m tensile strength at 2% and 5% strain respectively in both machine and cross-machine direction, junction efficiency not less than 90% of rib ultimate tensile strength and with 38mm X 38mm mesh opening.	sqm	97.00	Complete rate See the detailing of the item on chapter-A, Sl no. C-6
C-7	Supplying and laying of high radial stiffness and radial stiffness ratio polypropylene tri-axial punched and stretched geogrid with triangular aperture and hexagonal geometry conforming to ISO 10319/EOTA TR41 and with IRC accreditation for base/sub base stabilization in pavements with MIF value greater than 1.8 (IRC SP 59-2019-3.1.3 along with IRC : 37, 2018, clause no 7.2) for sub grade CBR≤3% with radial secant stiffness of 390 KN/m (tolerance 75 KN/m) at 5% strength, (MoRT&H's 5th Revision Table 700-7) and radial secant stiffness ratio of 0.7 to 0.8, junction efficiency greater than 90% with distance between alternate parallel ribs as 80mm.	sqm	110.30	Complete rate See the detailing of the item on chapter-A, Sl no. C-7
C-8	Supplying and laying of high strength flexible geogrids (HSFG) as soil reinforcement / basal reinforcement as per MoRT&Hs 3100, 5th Revision and IRC 113, made of high tenacity polyester core with polyethylene coating with Minimum Long Term Design Strength (LTDS) of more than 50% of ultimate tensile strength at 30 degree Celsius corresponding to 12 % strain.			Complete rate See the detailing of the item on chapter-A, Sl no. C-8
i)	High strength geogrid of Ultimate Tensile strength of 500 kN/m	sqm	561.30	
ii)	High strength geogrid of Ultimate Tensile strength of 900 kN/m	sqm	936.40	
iii)	High strength geogrid of Ultimate Tensile strength of 1200 kN/m	sqm	1208.00	

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-9	Supplying and laying of Reinforced Geo composite made of Polypropylene Polymer made three dimensional matrix of minimum mass per unit area 450 grams/sqm extruded onto a mechanically woven double twisted hexagonal shaped steel wire Mesh and Mesh Type 6X8 having Zn coating , the composite have a combined mass per unit area and nominal thickness of 12mm; including fixing of U pins of 8mm diameter of depth 500mm at 1m c/c spacing, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials,labour etc. complete and as directed by Engineer - In - Charge.(Ref to MoRT&H's 5 th Revision Specification 706)	sqm		<i>Complete rate See the detailing of the item on chapter-A, Sl no. C-9</i>
i)	Double twisted hexagonal shaped steel wire Mesh having dia 2.2mm and the composite shall have a combined mass per unit area more than 1630 Grams / Sq.m. <i>Note : Use for slope erosion protection work.</i>	sqm	919.30	
ii)	Double twisted hexagonal shaped steel wire Mesh having dia 2.2/3.2mm and the composite shall have a combined mass per unit area more than 1920 Grams / Sq.m. <i>Note : Use for slope erosion protection work.</i>	sqm	1029.30	
C-10	Supplying and laying of Reinforced Geocomposite made of Polypropylene Polymer made three dimensional matrix of minimum mass per unit area 450 grams/sqm extruded onto a mechanically woven double twisted hexagonal shaped steel wire Mesh and Mesh Type 8X10 having Zn coating, the composite have a combined mass per unit area and nominal thickness of 12mm; including fixing of U pins of 8mm diameter of depth 500mm at 1m c/c spacing, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials,labour etc. complete and as directed by Engineer - In - Charge.(Ref to MoRT&H's 5 th Revision Specification 706)			<i>Complete rate See the detailing of the item on chapter-A, Sl no. C-10</i>
i)	Double twisted hexagonal shaped steel wire Mesh having dia 2.7mm and the composite shall have a combined mass per unit area more than 1850 Grams / Sq.m.	sqm	854.30	
ii)	Double twisted hexagonal shaped steel wire Mesh having dia 2.7/3.7mm and the composite shall have a combined mass per unit area more than 2130 Grams / Sq.m.	sqm	964.30	
C-11	Providing and laying of woven Geotextile on the prepared subgrade as the separator cum reinforcement with necessary overlap as per drawing. .(Ref to MoRT&H's 5 th Specification 700 & 2500)			<i>Complete Rate See the detailing of the item on chapter-A, Sl no. C-11</i>
i)	For Type-I (160gsm) woven Geotextile	sqm	51.30	
ii)	For Type-II (240gsm) woven Geotextile	sqm	56.30	
iii)	For Type-III (450gsm) woven Geotextile	sqm	61.30	

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-12	Providing and laying of fibre glass geogrid manufactured from high quality glass fibre strands and coated with a polymer modified bitumen as reinforcement to Asphalt overlay flexible pavements and Asphalt over distressed rigid P.C.C. . (Ref to MoRT&H's 5 th Specification 708)			Complete Rate See the detailing of the item on chapter-A, Sl no. C-12
i)	For Tensile strength-50 kN/m	sqm	86.30	
ii)	For Tensile strength-100 kN/m	sqm	116.30	
C-13	Providing and laying of Needle Punched and mechanically bonded non-woven Geotextile indigenously manufactured from high quality fibers on the prepared subgrade for Separation. (Ref to MoRT&H's 5 th Specification 702)			Complete Rate See the detailing of the item on chapter-A, Sl no. C-13
i)	For Type-I (150gsm) non woven Geotextile	sqm	39.30	
ii)	For Type-II (200gsm) non woven Geotextile	sqm	48.30	
iii)	For Type-III (450gsm) non woven Geotextile	sqm	56.30	
C-14	Supplying and laying of drainage composite for use behind walls having three dimensional composite with thermobonding a draining core having "W" configuration in extruded monofilaments of minimum thickness 6.9mm with one filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75mm and tensile strength of 8.0 kN/m that will be working as separation or protecting layer, geocomposite having in plane flow capacity of 2.60 L / (m.s) at hydraulic gradient of 1.0 at 20 kpa pressure and tensile strength of 12 kN/m with mass per unit area of 540 gsm supplied in the form of roll for easy transportation to site of work as per detailed specification all complete as per directions of Engineer in charge. Note: Lower gsm may be used in RE structure.	sqm	416.10	Complete Rate See the detailing of the item on chapter-A, Sl no. C-14 (Ref to CPWD SOR 2016, Sub Head 26 for New Technologies and Materials Code No 8960).
C-15	Supplying and laying of drainage composite for use behind walls, between two different fills, alongside drains of road, below concrete lining of canals etc. having three dimensional composite with thermobonding a draining core having "W" configuration in extruded monofilaments of minimum thickness 7.2mm, with two filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75mm and tensile strength of 8.0 kN/m that will be working as separation or protecting layer, geocomposite having in plane flow capacity of 2.3 L / (m.s) at hydraulic gradient of 1.0 & 20 kPa pressure and tensile strength of 18 kN/m , with mass per unit area of 740 gsm, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials, labour etc. complete and as directed by Engineer - In - Charge. Note: Higher gsm may be used in drainage composite at the time of pavement construction	sqm	565.70	Complete Rate See the detailing of the item on chapter-A, Sl no. C-15 (Ref to CPWD SOR 2016, Sub Head 26 for New Technologies and Materials Code No 8960).

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-16	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	2414.17	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-16
C-17	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm , supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	3862.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-17
C-18	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm , supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	2776.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-18

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-19	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	1901.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-19
C-20	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	3068.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-20
C-21	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm , supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	2192.70	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-21

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-22	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x1m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	1681.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-22
C-23	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.5m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm, supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	2728.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-23.
C-24	Providing & making of BIS certified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit) with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12 (D=100 mm with tolerance of ± 2%) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.70/3.70 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 mm , supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	1943.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-24.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-25	Providing & making of Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	2026.25	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-25.
C-26	Providing & making of Gabion structure of size(2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500 of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	3617.50	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-26.
C-27	Providing & making of Gabion structure of size (2x0.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500 of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+10% Al alloy+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	2424	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-27.
C-28	Providing & making of Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	1641.25	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-28.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-29	Providing & making of Gabion structure of size (2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	2952.50	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-29.
C-30	Providing & making of Gabion structure of size (2x.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc+Polymer coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	1969	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-30.
C-31	Providing & making of Gabion structure of size (2x1x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	1476.25	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-31.
C-32	Providing & making of Gabion structure of size (2x0.5x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	2667.50	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-32.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-33	Providing & making of Gabion structure of size (2x0.8x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 10x12(D=100 mm with tolerance of ±2%), Zinc coated, Mesh wire diameter 2.7/3.7mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2/3.2mm(ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per directions of Engineer-in-charge.	cum	1774	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-33.
C-34	Providing & making of BIS certified Gabion structure of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of ± 2%) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	6964	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-34.
C-35	Providing & making of BIS certified Gabion structure of size (2x0.23x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of ± 2%) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	5412.00	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-35.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-36	Providing & making of BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn + 10% Al alloy +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	5721	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-36.
C-37	Providing & making of BIS certified Gabion structure of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	5895	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-37.
C-38	Providing & making BIS certified Gabion structure of size (2x0.23x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	4558	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-38.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-39	Providing & making of BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn +Polymer coated, Mesh wire diameter 2.20/3.20 mm (ID/OD), mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	3674	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-39.
C-40	Providing & making of BIS certified Gabion structure of size (2x0.17x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh wire diameter 2.20 mm , mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	5437	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-40.
C-41	Providing & making of BIS certified Gabion structure of size (2x0.23x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh wire diameter 2.20 mm , mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	4192	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-41.
C-42	Providing & making of BIS certified Gabion structure of size (2x0.3x1)m with Mechanically Woven Double Twisted Hexagonal Shaped Wire mesh Gabion mattress Boxes as per IS 16014:2012, MoRT&H's 5 th revision clause 2500, of required size, Mesh Type 6x8 (D=60 mm with tolerance of $\pm 2\%$) Zn coated, Mesh wire diameter 2.20 mm , mechanically edged/selvedged with partitions at every 1m interval and shall have minimum 10 numbers of openings per meter of mesh perpendicular to twist, tying with lacing wire of diameter 2.2 /3.2 mm (ID/OD), supplied @3% by weight of Gabion boxes, filled with boulders with least dimension of 200 mm, as per drawing, all complete as per direction of Engineer-in-charge.	cum	3369	Cost of material to be added. Consumption of material i.e. for boulder to be followed as per chapter A Sl No C-42.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
C-43	Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn + 10% Al alloy +Polymer Coated wire mesh, Mesh Size10x12. The approximate minimum weight of the sack gabion would be 1350 kgs when filled as per detailed technical specification.	cum	2102	Cost of material to be added. Consumption of material i.e. for sand to be followed as per chapter A Sl No C-43.
C-44	Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn +Polymer Coated wire mesh, Mesh Size10x12. The approximate minimum weight of the sack gabion would be 1350 kgs when filled as per detailed technical specification.	cum	1738	Cost of material to be added. Consumption of material i.e. for sand to be followed as per chapter A Sl No C-44.
C-45	Supply of Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter. The Sack Gabions shall be made of the mechanically woven hexagonal double twisted Zn Coated wire mesh, Mesh Size10x12. The approximate minimum weight of the sack gabion would be 1350 kgs when filled as per detailed technical specification.	cum	1582.30	Cost of material to be added. Consumption of material i.e. for sand to be followed as per chapter A Sl No C-45.

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
D) <u>New items related to traffic signs, marking and other road appurtenances Chapter 8 of NH SOR 19-20</u>				
D-1	Raised Profile Edge Markings with Reflectorised Hot Applied Thermoplastic Compound on Bituminous Road surfaces. Providing a laying of Raised Profile Edge Markings with Hot Applied Thermoplastic Road Markings along with Drop on Glass Beads as per the Provisions laid in IRC 35-2015, the height of the rib should be 8 mm for normal roads and it may extend upto 11 mm incase of expressway, the general gap between each rib shall vary between 200-500 mm, as per the direction of the Engineer in charge. Note : For Application on Rigid pavement a Resin Based primer has to be applied prior to the application of Thermoplastic Road Marking for which an additional cost of Rs.150 per Sqm shall be paid extra.	sqm	1205.00	<i>Complete Rate.</i> <i>Ribbed edge line marker as per Ministry's Circular.Ref Chapter A, D-1.</i>
D-2	Hybrid Solar-Retro Reflective Chevron Signs Supplying and Fixing of Solar Powered Retro Reflective LED Blinking Chevron signs as per the following specifications : Solar panel: 450 MaH/9v/4 watts, Battery: Li-ion 6 Volt 4.5 AH.C-10, Back up time: 108 hrs, Led: 112-130 nos, Display thickness : ACP or Alluminium 2 mm thick, Powder coated body. The Unit shall be self sufficient and shall be mounted on a Circular Pole: 50 NB (medium) duly painted and grouted into the soil using M15 concrete (45 cm x 45 cm x 60 cm) Size: 600 mm x 450 mm. The Retro reflective sheeting shall be HIP, Type IV as per IRC 67.	each	18000.00	<i>Complete Rate.</i> <i>See the detailing of the item on chapter-A, Sl no. D-2</i>
D-3	Solar Powered Road Studs Providing and fixing of solar powered road studs rectangular/circular shape (with super bright LEDs for long visibility from distance greater than 800m) having twin shank and body made of alluminium alloy or polycarbonate material which shall be absolutely weather resistant and strong enough to support a load of 13635 kg tested in accordance with ASTM-D 4280 and water resistance as per IP65 in reference with IS 12063:1987, fixed to the road surface using adhesives and the procedure recommended by manufacturer as per clause no 804.6 and 804.7 of MoRT&H's specification.	sqm	2159.00	<i>Complete Rate.</i> <i>See the detailing of the item on chapter-A, Sl no. D-3</i>

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
D-4	Wire Rope Safety Crash Barrier Supply and Installation of Wire Rope Safety Barrier, meeting and conforming to IRC:119:2015, Annexure II, Sec 4. The barrier shall be supplied complete with steel posts (spacing 3.0m c/c), Post length 1200mm, Post Cross Section not less than 100 x 50 x 16 mm, thickness 4 mm, plastic cap for post with retro-reflective sheet, plastic cable spacers, Stainless Steel Strap, rectangular steel sections having thickness 3mm and length 420mm to be used as post foundation sleeves, plastic top covers and bottom caps for steel sleeves, 3 galvanized wire ropes running along the entire length, each wire rope 19mm dia, (3x7)-pre-stretched-16.7 Ton capacity, required steel fittings, steel turnbuckles, steel anchors and M30 concrete complete. All steel components of the barrier shall be hot dip galvanized, having an average zinc coating thickness not less than 75 microns. The installation shall include excavating in soft soil for post foundations (250 mm dia x 610 mm depth at 3m c/c), excavation in all kind of soil for end anchor foundations (1m x 1m x1.5m each, 2 for each section), grouting of steel sleeves up to depth of 420 mm in each post foundation with M30 concrete, grouting anchor plates in each end anchor foundation with M30 concrete, placing of posts in the grouted steel sleeves ensuring that the posts and barrier are in line and level. Note : The bidder shall enclose all relevant papers in support of meeting the above criteria, the bidder shall also enclose Authorization from the OEM.	rm	3500.00	<i>Complete Rate. See the detailing of the item on chapter-A, Sl no. D-4</i>
D-5	Reflective Pavement Marker with Shank Providing and fixing of single mould twin shanked moulded road markers/ road studs made of poly carbonate or ABS body and shall support a load of 13635 kg tested in accordance to ASTM D4280 and complying with specifications as per clause 804 of MoRT&Hs 5th revision and section 5 of IRC-35:2015, the reflecting panels shall consist of number of lenses containing single or dual prismatic cubes capable of providing total internal reflection of the light entering the lenses face. The slope of retro reflecting surface should be 25 +/-5 degree to base. The reflective marker shall be fixed on the road surface using the epoxy/adhesive and the procedure recommended by the manufacturer. No nails shall be used to affix the markers so that they do not pose safety hazard on the road.	each	355.00	<i>Complete Rate. See the detailing of the item on chapter-A, Sl no. D-5</i>
D-6	Median Marker Made up of tough, impact resistant thermoplastic body, reflectorized on both side using approximate. 85 x 85 mm (for square shape or equivalent in case of round shape Fluorescent yellow prismatic sheeting (type-XI) reflective surface should be incline to approximate 20 degree to vertical to provide maximum reflective. Height & width of median marker shall be approximate by 110 mm complete with all fixing arrangements.	each	435.00	<i>Complete Rate See the detailing of the item on chapter-A, Sl no. D-6</i>

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
D-7	Retro-reflective studs Supplying and fixing of Metal Road studs with shank, made of Die-cast Aluminium alloy molded body with the size 100x100x19.8 mm, shank size 50 mm, having reflective panels of 34 glass reflective elements embedded in plastic and 19 degrees tilted to provide the highest reflectivity having reflective panels in yellow, red and white plastic with yellow, red, white or green glass elements. The weight of Aluminium stud should be not less than 370 grams. The stud shall have a compressive strength test certificate of 40 tons in accordance with ASTM D 4280-04, clause no 806 of 5 th revision of MoRT&Hs's and successfully passing retro reflectivity test as per EN 1463 part 1 & part 2 , from a reputed international laboratory. The fixing of road studs shall be done by using a recommended Two Component Epoxy after drilling 60 mm hole on the road for the shank to go inside.	each	485.00	Complete Rate See the detailing of the item on chapter-A, Sl no. D-7
D-8	Rumble Marking using two component cold plastic Providing and Laying of Rumble strips with 2 component cold plastic profiled material to give total width of 500 mm and height of 10 mm. Ramp provided to give width of 145 mm, width of ribs to be 55mm laid at centre to centre distance of 35 mm, with reflective glass beads @ 350 gm / sqm. The finished surface to be levelled, uniform and free from streaks and holes, to be applied on the edge lines, including secondary surface cleaning and cost of all materials etc. complete.	sqm	3888.00	Complete Rate See the detailing of the item on chapter-A, Sl no. D-8
D-9	Audible Vibratory Profile Marking with Hot Applied Thermoplastic and Reflectorizing Glass Providing and laying of raised profile edge line marking (audible vibratory) with special thermoplastic road marking compound 2 mm thick base coat layer above that ribs profile of 6 mm thick (total 8 mm thick) at the distance of 250 mm between two ribs including reflectorizing glass beads @ 250 gm / sq.m. area. The minium and maximum width of raised profile should be 100mm and 150 mm. The thickness of 8 mm profile should be exclusive of surface applied glass beads. The finished surface to be levelled, uniform and free from streaks and holes, to be applied on the edge lines, including secondary surface cleaning and cost of all materials etc. complete.	sqm	1012.00	Complete Rate See the detailing of the item on chapter-A, Sl no. D-9

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
<u>E) Steel Structure of Bridge</u>				
E-1	Plate Girder/Semi Through Girder Supplying of all types of structural steel conforming to Fe 410 B / Fe 490 of IS 2062, fabrication, assembling, erection/slewing/end launching of steel girders (not requiring traffic block) upto 45.7m span on sub structure including provision of trolley refuges, if required complete as per approved drawing including one coat of zinc chrome primer to IS:104& one coat Zinc Chrome Red Oxide to IS 2074 and painting as per IRC-B- 1 on all members (detailed fabrication and erection drawings & launching methodology will be prepared and got approved by the contractor from NH authority). Rate includes fabrication of all types of battens bracings, ties, stiffeners, packing diaphragms, shop rivets/ welding, T&F bolts,drafts, shop welds templates , jigs, fixtures, back up supports, accessories, transporting various components from fabrication shop to site including loading, unloading, lift and taxes complete, assembly of girders on drifts/bolts. field riveting/welding, assembling of temporary support for side slewing, raising of girders to the bed block level, providing sliding arrangements and slewing the girder in position, lowering of girder on bearings and bed plates grouting holes in the bed block for fixing of HD bolts/anchor pins of bed plates. All temporary arrangements, load testing may be done. The bearing sets to be provided with the girder will be paid separately .	MT	22581.54	<i>Cost of material to be added. Consumption of materials to be followed as per chapter A , Sl No. E-1</i>
E-2	Open WEB Girder Basic cost for fabrication,erection,leading, painting primer coats and final two coats of paint at site Take a Plate girder of 200 Feet length and 1955.3 Qtl weight and 4000 Sqm of area to be painted. Supplying and laying drainage composite wrapped in sand layer,having three dimensional composite with thermobonding a draining core having "W" configuration in extruded monofilaments of minimum thickness 7.2mm, with two filtering UV stabilized polypropylene nonwoven geotextile of minimum thickness of 0.75mm and tensile strength of 8.0 kN/m that will be working as separation or protecting layer, geocomposite having in plane flow capacity of 2.3 L / (m.s) at hydraulic gradient of 1.0 & 20 kPa pressure and tensile strength of 18 kN/m , with mass per unit area of 740 gsm, at easily accessible location including top and bottom, with all leads and lifts, manpower and machinery, materials, labour etc. complete and as directed by Engineer - In - Charge.	MT	29745.00	<i>Cost of material to be added. Consumption of materials to be followed as per chapter A , Sl No. E-2</i>
E-3	Metalizing of steel work of girder with grit blasting. Metalizing of steel work of girder with grit blasting ,following by one coat of each primer (as per IS:5666) one coat of Zink Chrome primer (as per IS: 104) and two coats of Aluminium paint (as per IS:2339) with all labour, T&P and material as a complete job	sq.m	391.00	<i>Complete rate See the detailing of the item on chapter-A, Sl no. E-3</i>

ADDENDA TO CHAPTER - 17

MISCELLANEOUS

Serial No.	Item	Unit	Rate (₹)	Remarks
			Labour, T&P, Machinery etc.	
1	2	3	4	5
E-4	Cleanning of steelwork of bridge girders with scrappers and / or ware brushes to remove all rust and loose or perished paint to proper perfectly clean and dry bare metal surface free from all dust/ foreign materials and ready for initial coat of paint. Rate includes cost of labour, consumables, brushes, tools and plants, labours, scaffolding, jhoola, hanging scaffolding staging etc.			Complete rate See the detailing of the item on chapter-A, Sl no. E-4
	a) For bridge girders upto 30.5 m clearspan	sq.m	31.00	
	b) For larger spans and heavier loads and extra height of scaffolding	sq.m	46.00	
E-5	Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required. With one coat of ready mix red lead conforming to IS 102			Complete rate See the detailing of the item on chapter-A, Sl no. E-5
	a) Painting Work of Bridge Girder upto 30.5 m	sq.m	33.40	
	b) Painting Work of Bridge Girder above 30.5 m	sq.m	37.20	
E-6	Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required. With one coat of ready mix Zinc Chromate conforming to IS 104 followed by one coat of Zinc Chromate red oxide conforming to IS:2074			Complete rate See the detailing of the item on chapter-A, Sl no. E-6
	a) Painting Work of Bridge Girder upto 30.5 m	sq.m	52.20	
	b) Painting Work of Bridge Girder above 30.5 m	sq.m	58.30	
E-7	Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required. With two coats Alluminium paints in dual containers conforming to IS 2339			Complete rate See the detailing of the item on chapter-A, Sl no. E-7
	a) Painting Work of Bridge Girder upto 30.5 m	sq.m	65.20	
	b) Painting Work of Bridge Girder above 30.5 m	sq.m	71.20	
E-8	Painting cleaned bridge plate girder including all scaffolding shuttering strutting along with provision of jhoola / hanging scaffolding where required. With two coats epoxy paint			Complete rate See the detailing of the item on chapter-A, Sl no. E-8
	a) Painting Work of Bridge Girder upto 30.5 m	sq.m	92.50	
	b) Painting Work of Bridge Girder above 30.5 m	sq.m	98.60	

TABLES

COST OF PAVER BLOCK

TABLE - I (PART-C)

Serial No.	Description of Items	Unit	Kolkata/ 24 Parganas (N&S), Howrah, Hooghly, Nadia, Murshidabad, Malda, N&S Dinajpur, Burdwan, E&W Midnapore, Bankura, Purulia, Birbhum,	Coochbehar, Jalpaiguri, Alipurduar, Darjeeling Plains	Darjeeling Hill Area	Remarks
1	<u>Concrete Vibro Hydraulic Paver Block (Not less than M40)</u>					<i>Rate at site</i>
	(a) 120mm thick	sqm	1500	1750	1900	
	(b) 100mm thick	sqm	1250	1450	1575	
	(c) 80mm thick	sqm	950	1100	1195	
	(d) 60mm thick	sqm	800	935	1015	

Note : The above rates are exclusive of contractor's profit & GST.

ANNEXURE

Illustration for New Items related to Chapter 17 of NHSOR 2019-20**A) New Items related to Bases and Surface courses (Bituminous) Chapter 5****A-1. Illustration for analysis of rate for Mechanised Mastic Ashphalt (25mm thickness)**(Ref. Sr. No. A-1 of Chapter - 17, Page - 1 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

For this illustration a site 15 km away from Chitpur railway stackyard (Kolkata) has been considered.

Distance from Chitpur (Kolkata) Railway Stackyard to worksite = 15 Km.

Lead by road = 15 Km.

i) Cost of carriage : (Plain Area) Unit : cum.

From Km.	To Km.	Rate @ (₹)	Distance	Amount (₹)	Remarks
0	15	11.10	15	166.50	For Carriage cost analysis refer Chapter-I, Page`-68, of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June' 2019).
Total carriage cost = ₹				166.50	per cum.
Carriage cost =				166.5	per cum.

ii) Cost of Pakur variety stone aggregate at worksite :

Item	Size	Cost of Pakur variety material at Chitpur (Kolkata) Rly Yard per cu.m.	Loading & Unloading	Carriage Cost	Total Cost	REMARKS
	(mm)	(₹)	(₹)	(₹)	(₹)	
Ballast	75	1514	125	166.50	1805.5	1) For Cost of Pakur variety material at Chitpur (Kolkata) Rly Yard. refer Table-VI, page-280 of N.H. Schedule of Rates,2019-2020 (w.e.f 1st June ' 2019). This rate excludes contractor's profit. 2) For loading/unloading cost refer Chapter-I, Page- 68 of N.H. Schedule of Rates,,2019-2020 (w.e.f 1st June '2019) 3) Material costs are in loose net volume
Ballast	63	1714	125	166.50	2005.5	
Ballast	53	1784	125	166.50	2075.5	
Ballast	45	1784	125	166.50	2075.5	
Ballast	40	1844	125	166.50	2135.5	
Ballast	37.5	1844	125	166.50	2135.5	
Chips	26.5	1804	125	166.50	2095.5	
Chips	20/22.4	1894	125	166.50	2185.5	
Chips	13.2	1914	125	166.50	2205.5	
Chips	10/11.2	1564	125	166.50	1855.5	
Chips	5.6	1234	125	166.50	1525.5	
Dust	-	1214	125	166.50	1505.5	

iii) Cost of Bulk Bitumen 85/25 at site

Source of material :- Haldia Refinery

Lead from Haldia Refinery to Kolkata worksite = 120 Km.

COST OF ROAD CARRIAGE : (Plain Area) Unit : tn

From Km.	To Km.	Rate @ (₹)	Distance	Amount (₹)	REMARKS
0	120	7	120	840.00	For Carriage cost analysis refer Chapter-I, Page-69,of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June' 2019).
Total carriage cost = ₹				840.00	per tonne

Carriage cost = ₹ 840 per tonne

Cost of Bitumen (85/25) at Source (Excluding all Taxes) ₹ 39,910.00

Loading and unloading ₹ 169.00

Carriage Cost ₹ 840.00

₹ 40,919.00 per tonne**iv) Cost of Bulk Bitumen VG 30 at site**

Cost of Bitumen at worksite ₹ 32,910.00

Loading and unloading ₹ 169.00

Carriage Cost ₹ 840.00

Cost of Bitumen at worksite ₹ 33,919.00 per tonne

Rate for Mechanised Mastic Asphalt (25mm thickness)

Unit : sq.m.

A. Labour, T & P, Machinery etc.

200.00

B. Materials

329.26

Item	Unit	Quantity	Rate (₹)	Total Cost(₹)	REMARKS
Bitumen 85/25	kg	5.714	40.919	233.81	Bitumen Price (85/25)=32910+7000=39910 per MT, (VG 30 + 7000) is the published rate of I.O.C on 1st October', 2019 is considered, i.e. on the 4th quarter of the financial year 2019-2020
Coarse Aggregate					
5.6mm (50% of vol. of total	cum	0.0079	1525.50	11.98	
13.2 mm (50% of vol. of total	cum	0.0079	2205.50	17.31	
sand	cum	0.011	1180.00	12.98	
Lime stonedust with calcium	kg	10.29	5.00	51.45	
13.2mm stone metal for	cum	0.00057	2205.50	1.26	
Bitumen VG-30	kg	0.014	33.919	0.47	
Material Cost				329.26	

(A+B) ₹ 529.26

C. Cost of formwork/scaffolding etc.@ 0% of B ₹ 0.00

Cost of (A+B+C) ₹ 529.26

D. Add overhead charges @ 10 10% of (A+B+C) ₹ 52.93

E. Add contractor's profit @ 10% of (A+B+C+D) ₹ 58.22

TOTAL COST (A+B+C+D+E) ₹ 640.41

F. Add Labour welfare cess @ 1% of (A+B+C+D+E) ₹ 6.40

ITEM RATE ₹ 646.81

A-2. Illustration for analysis of rate for Waste plastic Modified Dense Graded Hot Bituminous Mix for Wearing Coarse.

(Ref. Sr. No. A-2, of Chapter - 17, Page - 1 of 2nd Addenda of N.H. Schedule of Rates 2019-2020

For this illustration a site has been considered **at Burdwan District**

Distance from Burdwan Railway Stackyard to worksite 15 Km.

Lead by road = 15 Km.

Cost of Road Carriage : (Plain Area) Unit : cum.

From Km.	To Km.	Rate @ (₹)	Distance	Amount (₹)	REMARKS
0	15	11.10	15	166.50	For Carriage cost analysis refer Chapter-I, Page -68, of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June' 2019).
Total carriage cost = ₹				166.50	

ii) Cost of Pakur variety stone aggregate at worksite :

Item	Size	Cost of Pakur variety material at Burdwan Jn. Rly Yard per cu.m.	Loading & Unloading	Carriage Cost	Total Cost	Remarks
	(mm)	(₹)	(₹)	(₹)	(₹)	
Ballast	75	1257	125	166.50	1548.5	1) For Cost of Pakur variety material at Burdwan Jn. Rly Yard. refer Table-VI, page-280 of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June ' 2019). This rate excludes contractor's profit. 2) For loading/unloading cost refer Chapter-I, Page- 68 of N.H. Schedule of Rates,, 2019-2020 (w.e.f 1st June '2019) 3) Material costs are in loose net volume
Ballast	63	1457	125	166.50	1748.5	
Ballast	53	1527	125	166.50	1818.5	
Ballast	45	1527	125	166.50	1818.5	
Ballast	40	1587	125	166.50	1878.5	
Ballast	37.5	1587	125	166.50	1878.5	
Chips	26.5	1547	125	166.50	1838.5	
Chips	20/22.4	1637	125	166.50	1928.5	
Chips	13.2	1657	125	166.50	1948.5	
Chips	10/11.2	1307	125	166.50	1598.5	
Chips	5.6	977	125	166.50	1268.5	
Dust	-	957	125	166.50	1248.5	

Unit : cu.m.
Source of material :- Haldia Refinery
Lead from Haldia Refinery to Burdwan worksite = 197 Km.

Cost of Road Carriage : (Plain Area) Unit : tn

From Km.	To Km.	Rate @ (₹)	Distance	Amount (₹)	REMARKS
0	197	7	197	1379.00	For Carriage cost analysis refer Chapter-I, Page-69, of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June' 2019).
Total carriage cost = ₹				1379.00	per tonne

Cost of Bulk Bitumen VG 30 at site

Cost of Bitumen at worksite ₹ 32,910.00
Loading and unloading ₹ 169.00
Carriage Cost ₹ 1,379.00
Cost of Bitumen at worksite ₹ 34,458.00 per tonne

Rate of Waste Plastic

A. Labour, T & P, Machinery etc. 1561.60
B. Materials 6598.99

Item	Unit	Quantity	Rate	Total	Remarks
			(₹)	(₹)	Bitumen may be VG 30 or VG 40 as per design mix depending on location of site and volume of traffic. Bitumen Price VG 30 per MT, is the published rate of I.O.C on 1st October '2019 is considered, i.e. on the 4th quarter of the financial year 2019-2020
Bitumen VG-30	kg	127	34.46	4376.42	
Coarse Aggregate					
13.2 mm	cum	0.0700	1948.50	136.40	
10.00 mm	cum	0.2400	1598.50	383.64	
5.6 mm	cum	0.2500	1268.50	317.13	
Stone dust	cum	0.9000	1248.50	1123.65	
cement as filler material	kg	44.57	5.87	261.76	
Material Cost				6598.99	

(A+B) ₹ 8160.59
C. Cost of formwork/scaffolding etc.@ 0% of B ₹ 0.00
Cost of (A+B+C) ₹ 8160.59
D. Add overhead charges @ 10% of (A+B+C) ₹ 816.06
E. Add contractor's profit @ 10% of (A+B+C+D) ₹ 897.66
TOTAL COST (A+B+C+D+E) ₹ 9874.31
F. Add Labour welfare cess @ 1% of (A+B+C+D+E) ₹ 98.74
ITEM RATE ₹ 9973.06

B) New Items related to Cement Concrete Pavement Chapter 6

B-1. Illustration for analysis of rate for White Topping

(Ref. Sr. No.B-1 of Chapter - 17, Page - 2 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

(For Midnapore District)

For this illustration a site 20 km away from Kharagpur railway stackyard has been considered.

Distance from Kharagpur Railway Stackyard to worksite 20 Km.

Lead by road = 20 Km.

Cost of Road Carriage : (Plain Area) Unit : cum.

From Km.	To Km.	Rate @ (₹)	Distance	Amount (₹)	REMARKS
0	20	11.10	20	222.00	For Carriage cost analysis refer Chapter-I, Page -68, of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June' 2019).
Total carriage cost = ₹				222.00	per cum.

Cost of Pakur variety stone aggregate at worksite :

Item	Size (mm)	Cost of Pakur variety material at Kharagpur Jn. (₹) Rly Yard per	Loading & Unloading (₹)	Carriage Cost (₹)	Total Cost (₹)	REMARKS
Boulder	225 to 150	1817	125	222.00	2164	1) For Cost of Pakur variety material at Kharagpur Rly Yard. refer Table-VI, page-280 of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June '2019). This rate excludes contractor's profit. 2) For loading/unloading cost refer Chapter-I, Page- 68 of N.H. Schedule of Rates,, 2019-2020 (w.e.f 1st June '2019) 3) Material costs are in loose net volume
Ballast	75	1767	125	222.00	2114.0	
Ballast	63	1967	125	222.00	2314.0	
Ballast	53	2037	125	222.00	2384.0	
Ballast	45	2037	125	222.00	2384.0	
Ballast	40	2097	125	222.00	2444.0	
Ballast	37.5	2097	125	222.00	2444.0	
Chips	26.5	2057	125	222.00	2404.0	
Chips	20/22.4	2147	125	222.00	2494.0	
Chips	13.2	2167	125	222.00	2514.0	
Chips	10/11.2	1817	125	222.00	2164.0	
Chips	5.6	1487	125	222.00	1834.0	
Dust	-	1467	125	222.00	1814.0	

Unit : cu.m.

Item Rate for White Topping

A. Labour, T & P, Machinery etc.

1645.20

B. Materials

4984.20

Item Rate for White Topping	Unit	Quantity	Rate	Total Cost	REMARKS
			(₹)	(₹)	
Stone Aggregate					
26.5mm	cum	0.54	2404.00	1298.16	
11.2mm		0.36	2164.00	779.04	
cement	tn	0.40	6056.00	2422.40	
sand	cum	0.45	798.00	359.10	
Steel for Dowel	tn	0.00220	57054.00	125.50	
Material Cost				4984.20	

(A+B) ₹ 6629.40

C. Cost of formwork/scaffolding etc.@ 0%

of B ₹ 0.00

Cost of (A+B+C) ₹ 6629.40

D. Add overhead charges @

10 10% of (A+B+C) ₹ 662.94

E. Add contractor's profit @

10% of (A+B+C+D) ₹ 729.23

TOTAL COST (A+B+C+D+E) ₹ 8021.58

F. Add Labour welfare cess

@ 1% of (A+B+C+D+E) ₹ 80.22

ITEM RATE ₹ 8101.80

B-3. Illustration for analysis of rate for Concrete Block Pavement

(Ref. Sr. No. B-3 of Chapter - 17, Page - 3 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

For this illustration a site has been considered at Burdwan District

Unit : sq.m.

Type I - 120mm thickness

A. Labour, T & P, Machinery etc.

151.00

B. Materials

1620.85

ITEM	UNIT	Quantity	Rate	TOTAL COST	REMARKS
			(₹)	(₹)	
Cost of Paver Block	sq.m	1.05	1500.000	1575.00	For the cost of paver block at the Burdwan Dist.Ref Table No.1, Part B
sand					
Bedding Sand (80% of 0.06 m3)	cum	0.05	798.00	39.90	
Joint filling sand (20% of 0.06m3)	cum	0.01	595.00	5.95	
Material Cost				1620.85	

				(A+B) ₹	1771.85
C. Cost of formwork/scaffolding etc.@ 0%		of B	₹		0.00
Cost of (A+B+C)				₹	1771.85
D. Add overhead charges @ 10 10%	of	(A+B+C)	₹		177.19
E. Add contractor's profit @ 10%	of	(A+B+C+D)	₹		194.90
TOTAL COST (A+B+C+D+E)				₹	2143.94
F. Add Labour welfare cess @ 1%	of	(A+B+C+D+E)	₹		21.44
ITEM RATE				₹	2165.38
					2165.50

C) New Items related to Geosynthetic and Reinforced earth Chapter 7**C-16. Illustration for Analysis of rate for BIS cerified Gabion fascia box with 2m integrated tail of Size 3m(Length of Tail)x3m(Length of Box)x0.8m(Height of unit)**

(Ref. Sr. No. C-16 of Chapter - 17, Page - 8 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

Unit : cu.m.

A. Labour, T & P, Machinery etc.

2414.00

B. Materials

2164.00

Item	Unit	Quantity	Rate (₹)	Total Cost (₹)	REMARKS
Boulder 200mm size	cum	1.00	2164.000	2164.00	Consider the work site 15 km from Kharagpur railway stackyard (Pakur Variety)
Material Cost				2164.00	

				(A+B) ₹	4578.00
C. Cost of formwork/scaffolding etc.@ 0%		of B	₹		0.00
Cost of (A+B+C)				₹	4578.00
D. Add overhead charges @ 10 10%	of	(A+B+C)	₹		457.80
E. Add contractor's profit @ 10%	of	(A+B+C+D)	₹		503.58
TOTAL COST (A+B+C+D+E)				₹	5539.38
F. Add Labour welfare cess @ 1%	of	(A+B+C+D+E)	₹		55.39
ITEM RATE				₹	5594.77

C-45. Illustration for Analysis of rate for Cylindrical Gabions /Sack Gabions of size 2 m in length and 0.96 m diameter.

(Ref. Sr. No. C-45 of Chapter - 17, Page - 16 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

Unit : cu.m.

A. Labour, T & P, Machinery etc. 1582.30

B. Materials 625.00

Item	Unit	Quantity	Rate (₹)	Total Cost (₹)	Remarks
Sand	cum	1.00	625.000	625.00	Consider the work site at Midnapore District
Material Cost				625.00	

(A+B) ₹ 2207.30

C. Cost of formwork/scaffolding etc.@ 0% of B ₹ 0.00

Cost of (A+B+C) ₹ 2207.30

D. Add overhead charges @ 10% of (A+B+C) ₹ 220.73

E. Add contractor's profit @ 10% of (A+B+C+D) ₹ 242.80

TOTAL COST (A+B+C+D+E) ₹ 2670.83

F. Add Labour welfare cess @ 1% of (A+B+C+D+E) ₹ 26.71

ITEM RATE ₹ 2697.54

E) New Items related to Steel Structure of Bridge

E-1. Illustration for analysis of rate for Bridgework - Super structure - Steel Plate Girder/Semi Through Girder

(Ref. Sr. No. E-1, of Chapter - 17, Page - 20 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

(For Burdwan District)

Unit : Tonne

Supply of HYSD reinforcement bars shall be from Market at Burdwan

Total Lead from Site 15 km

Cost of Road Carriage : (Plain Area) Unit : cum.

From Km.	To Km.	Rate @ (₹)	Distance (Km.)	Amount (₹)	REMARKS
0	15	7	15	105.00	For Carriage cost analysis refer Chapter-I, Page-69, of N.H. Schedule of Rates, 2019-2020 (w.e.f 1st June ' 2019)..
Total carriage cost = ₹				105.00	per tonne

Cost of Steel at worksite

Item	Unit	Basic Cost (₹)	Loading & Unloading (₹)	Carriage Cost (₹)	Total Cost (₹)	REMARKS
Mild steel plates	tonne	49205	169.00	105.00	49479.00	Refer values in Annexure - I, page 283 of N.H. Schedule of Rates 2019-2020 (w.e.f 1st June' 2019)..
Rivets	tonne	56888	169.00	105.00	57162.00	
20mm dia holding down bolts	tonne	56888	169.00	105.00	57162.00	

A. Labour, T & P, Machinery etc.

22581.54

B. Materials

53791.11

Item	Unit	Quantity	Rate (₹)	Total Cost (₹)	REMARKS
Mild steel plates	MT	1.05	49479.00	51952.95	
Rivets	MT	0.0315	57162.00	1800.60	
20mm dia holding down bolts	MT	0.000657	57162.00	37.56	
Material Cost				53791.11	

(A+B) ₹ 76372.65

C. Add for sundries , mics, T & P, consumable etc.@ 3% of B ₹ 1613.73

Cost of (A+B+C) ₹ 77986.38

D. Add overhead charges @ 10 25% of (A+B+C) ₹ 19496.60

E. Add contractor's profit @ 10% of (A+B+C+D) ₹ 9748.30

TOTAL COST (A+B+C+D+E) ₹ 107231.27

F. Add Labour welfare cess @ 1% of (A+B+C+D+E) ₹ 1072.31

ITEM RATE ₹ 108303.59

E-2 Illustration for analysis of rate for Open WEB Grider.

(Ref. Sr. No. E-2, of Chapter - 17, Page - 20 of 2nd Addenda of N.H. Schedule of Rates 2019-2020)

A. Labour, T & P, Machinery etc.

29745.00

B. Materials

53760.81

Item	Unit	Quantity	Rate (₹)	Total Cost (₹)	REMARKS
Mild steel plates	MT	1.05	49479.00	51952.95	
Rivets	MT	0.0315	57162.00	1800.60	
20mm dia holding down bolts	MT	0.000127	57162.00	7.26	
Material Cost				53760.81	

(A+B) ₹ 83505.81

C. Add for sundries , mics, T & P, consumable etc.@ 3% of B ₹ 1612.82

Cost of (A+B+C) ₹ 85118.64

D. Add overhead charges @ 10 25% of (A+B+C) ₹ 21279.66

E. Add contractor's profit @ 10% of (A+B+C+D) ₹ 10639.83

TOTAL COST (A+B+C+D+E) ₹ 117038.13

F. Add Labour welfare cess @ 1% of (A+B+C+D+E) ₹ 1170.38

ITEM RATE ₹ 118208.51